

Qualifying Exam

November 7, 2025

Sam Buxbaum

Motivation

Secure computation has taken great leaps in recent years...

but it's still impractical in many scenarios.

Goal: Fill in the gaps that prevent the use of MPC in practice

Scaling Up MPC

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2. Directly incorporating more parties

- Significant practical challenges
- Distribute trust more widely

Oblivious Shuffle Preprocessing

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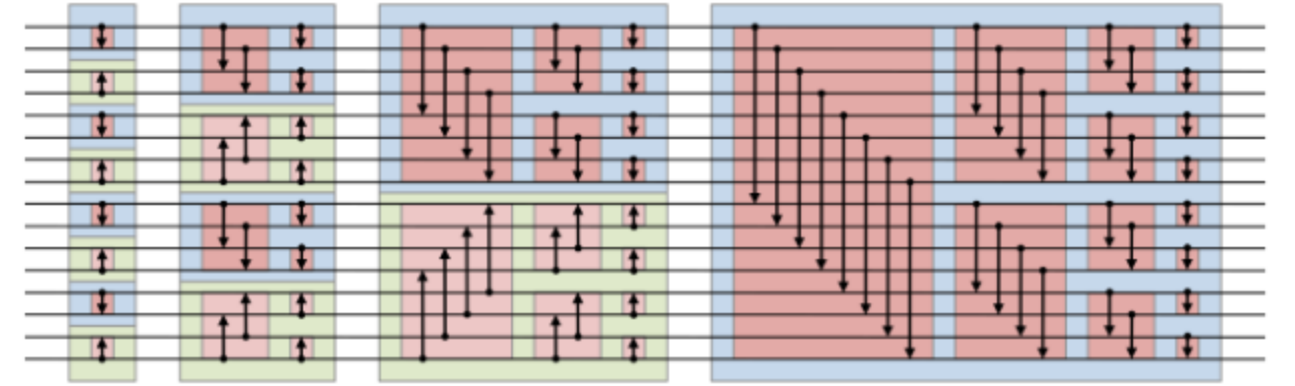
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- The output is a secret sharing $[\pi(\mathbf{x})]$

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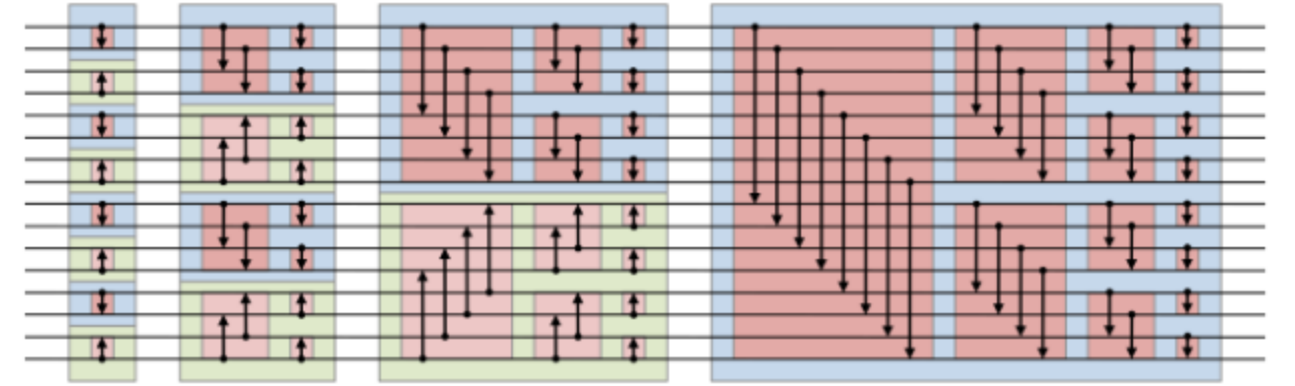
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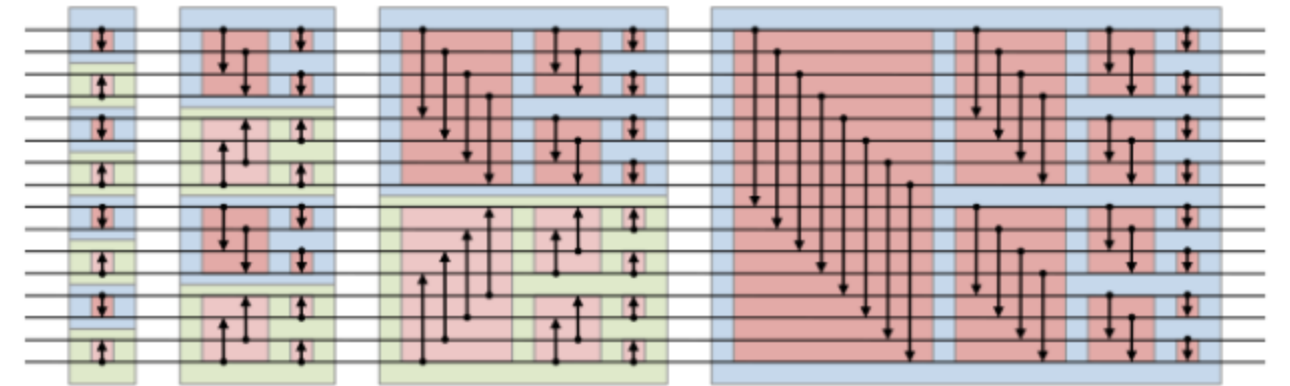


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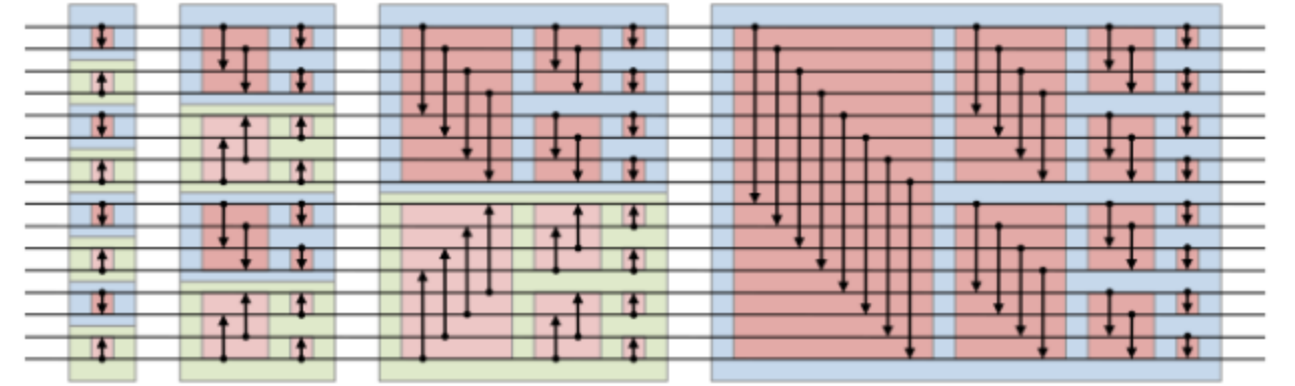
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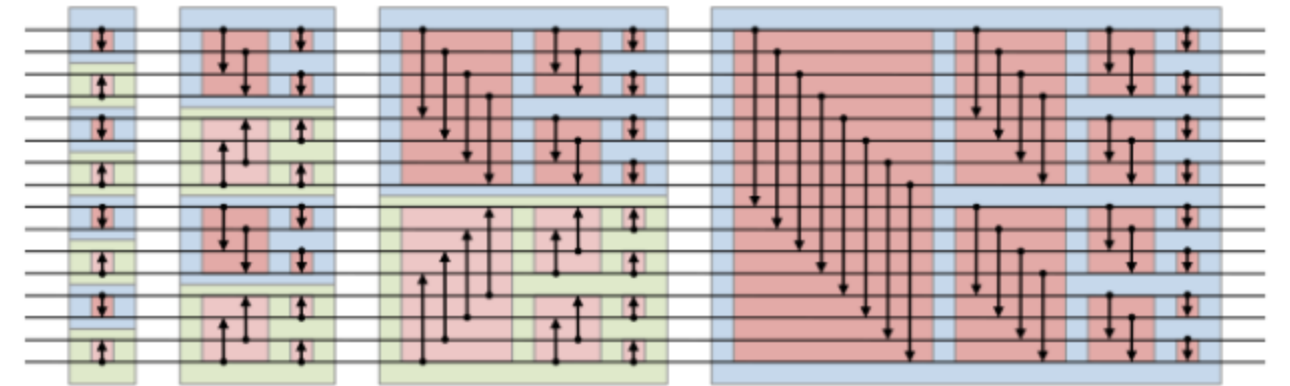
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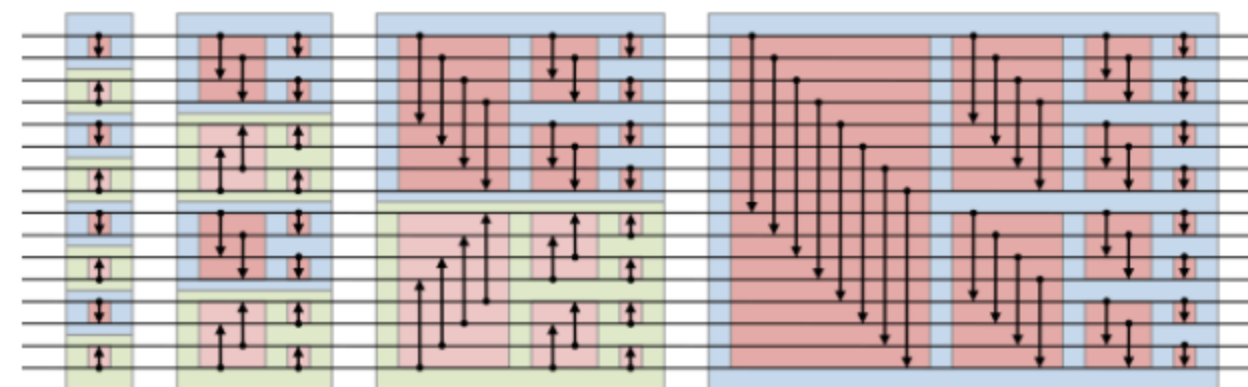
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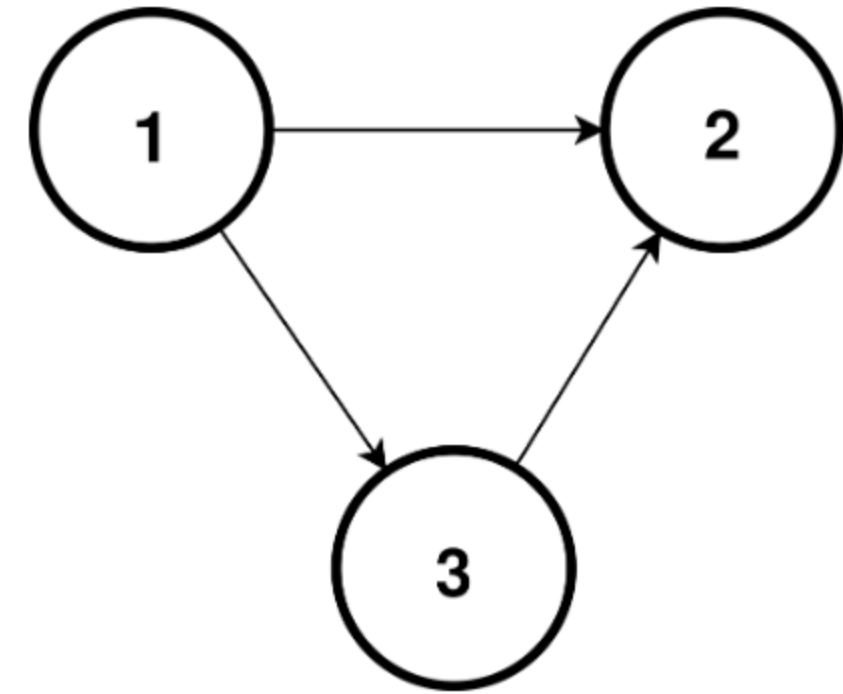
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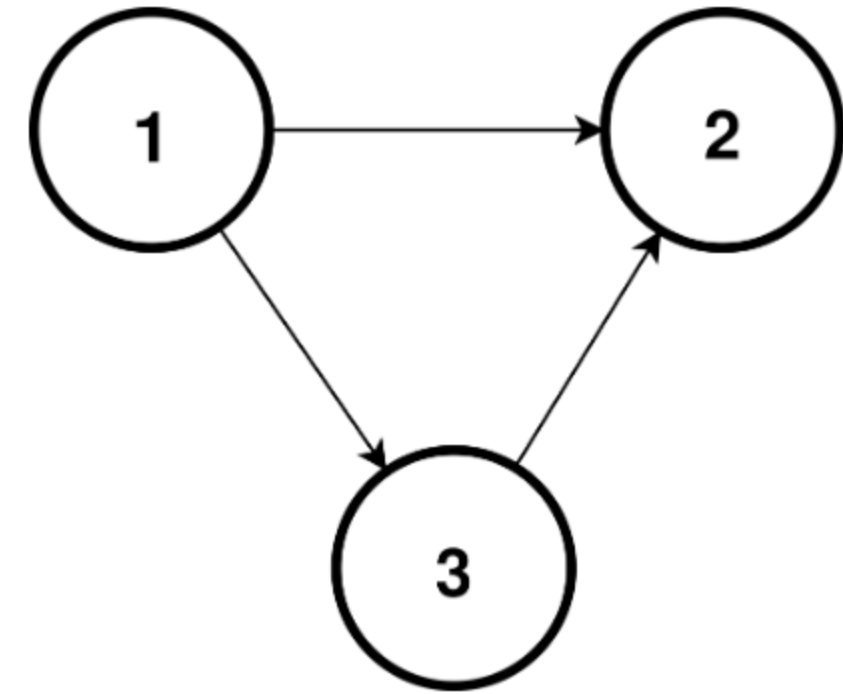
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- Achieves $O(n \log n)$ secure comparisons

Example: Graph Algorithms



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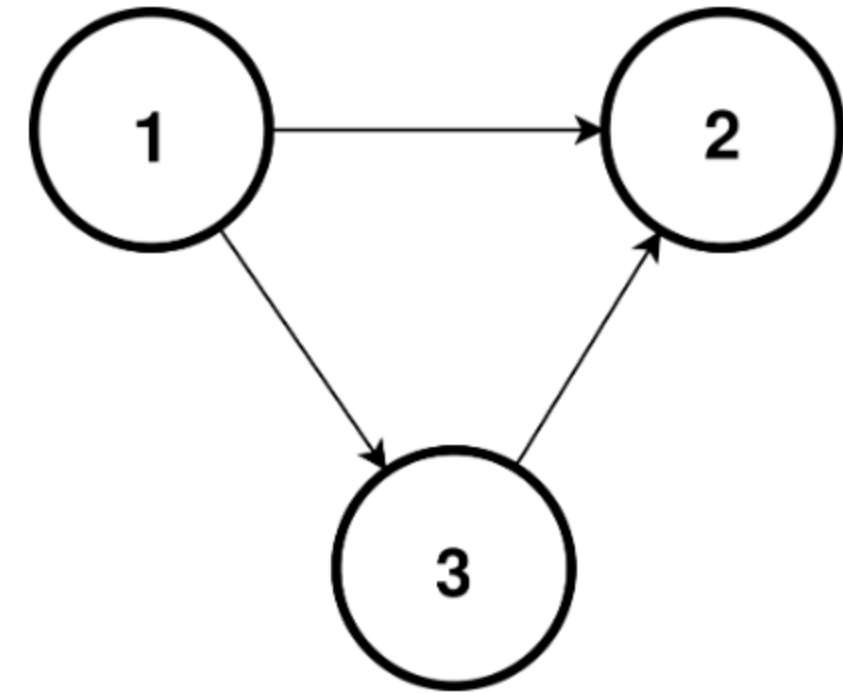
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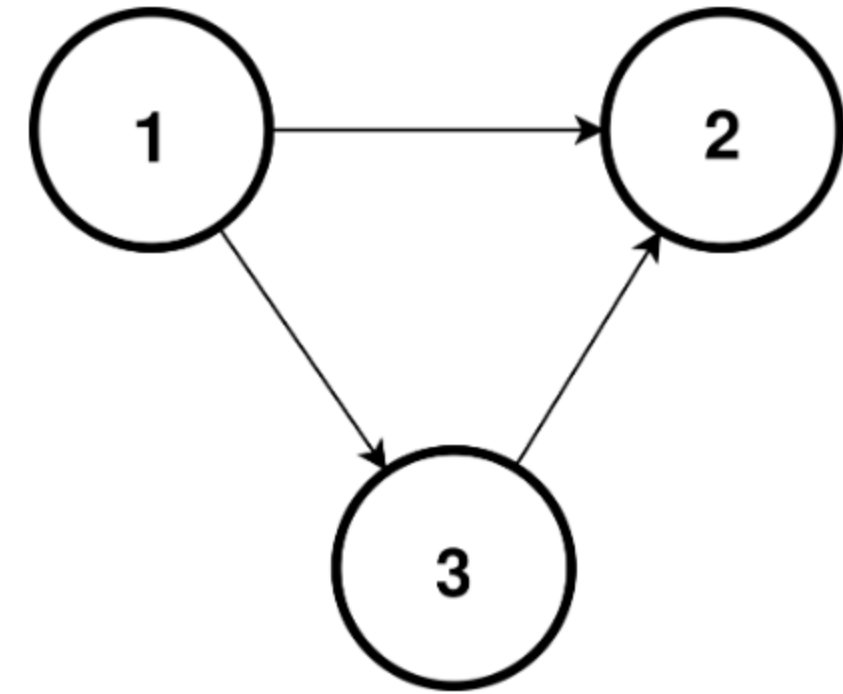


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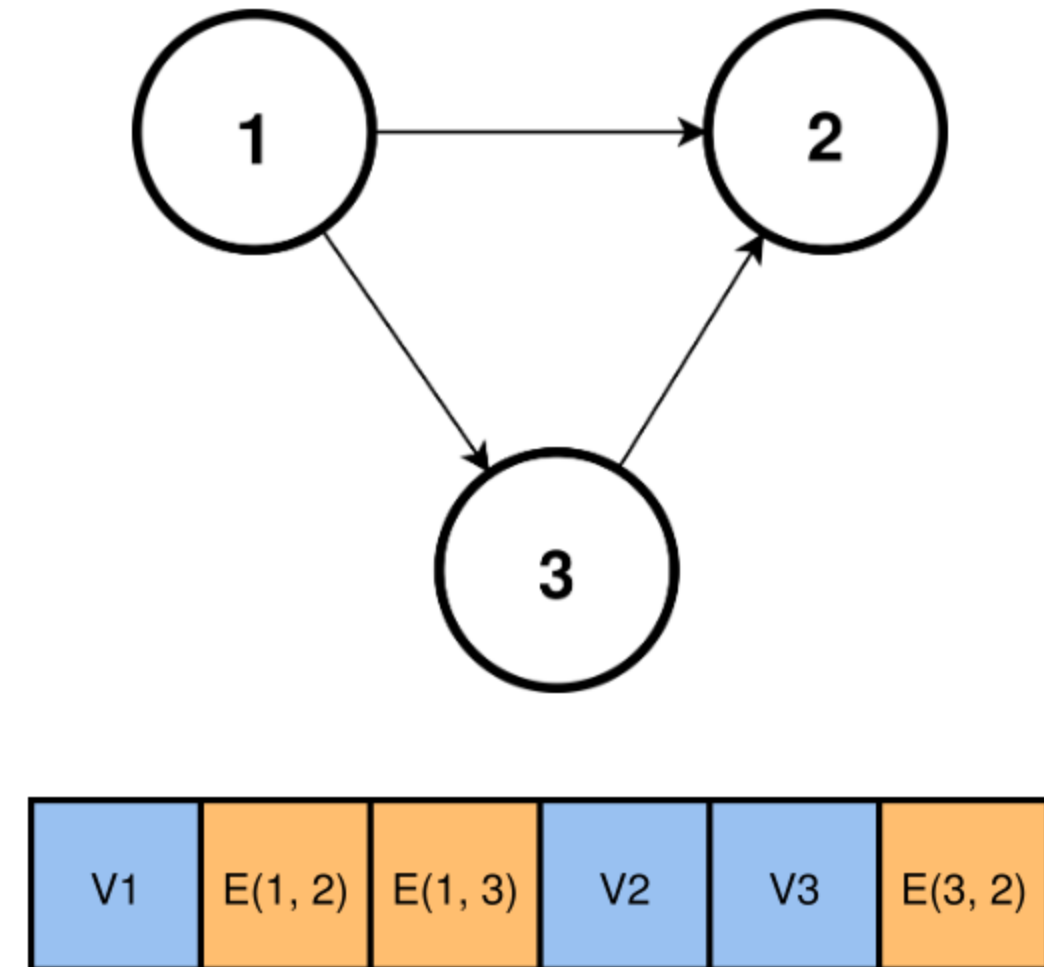
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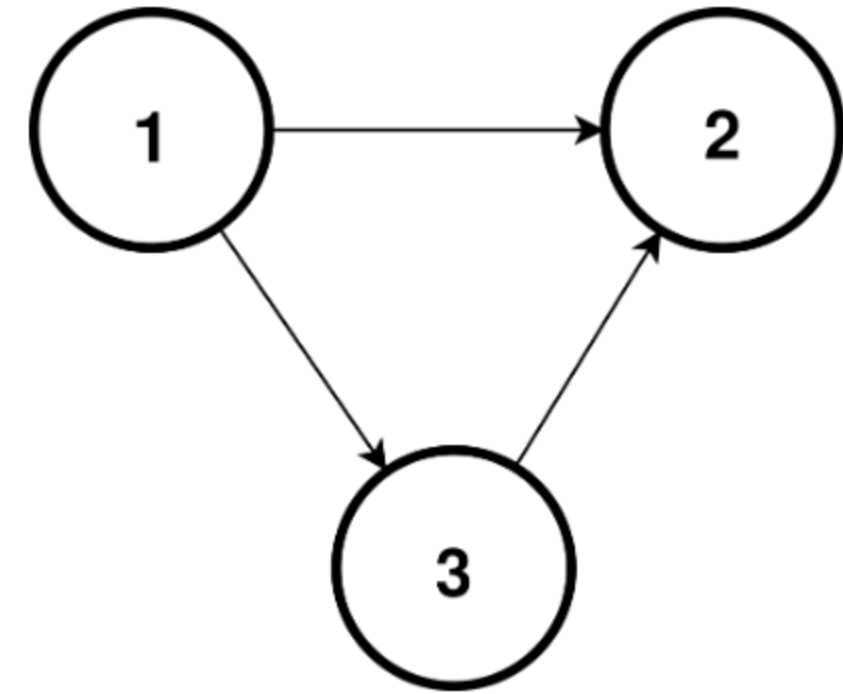
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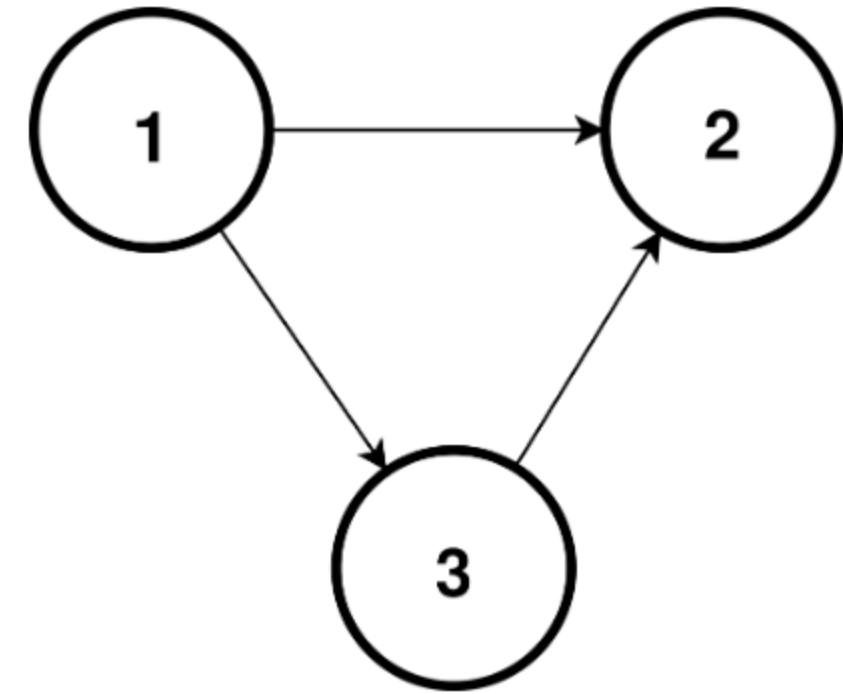
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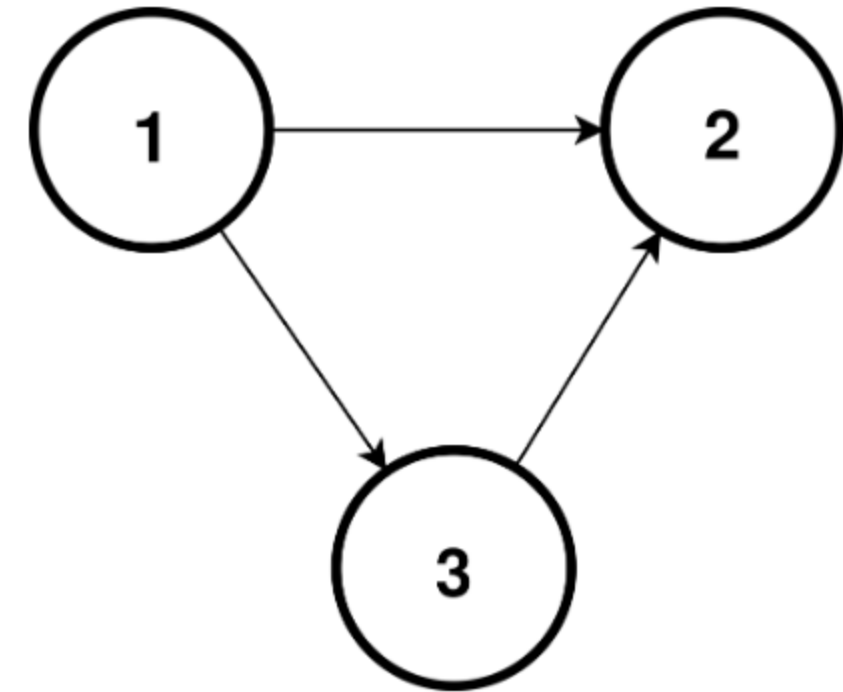
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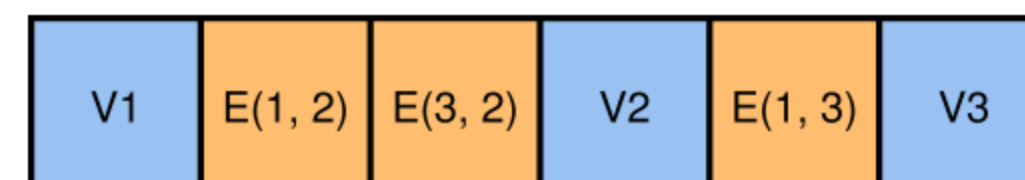
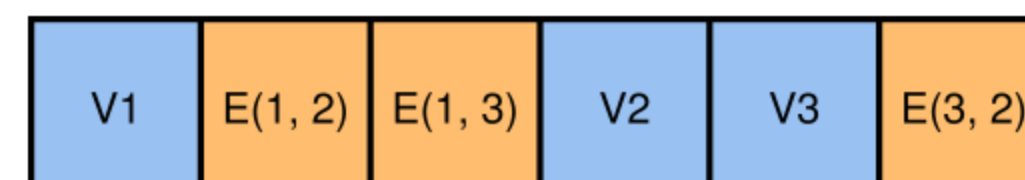
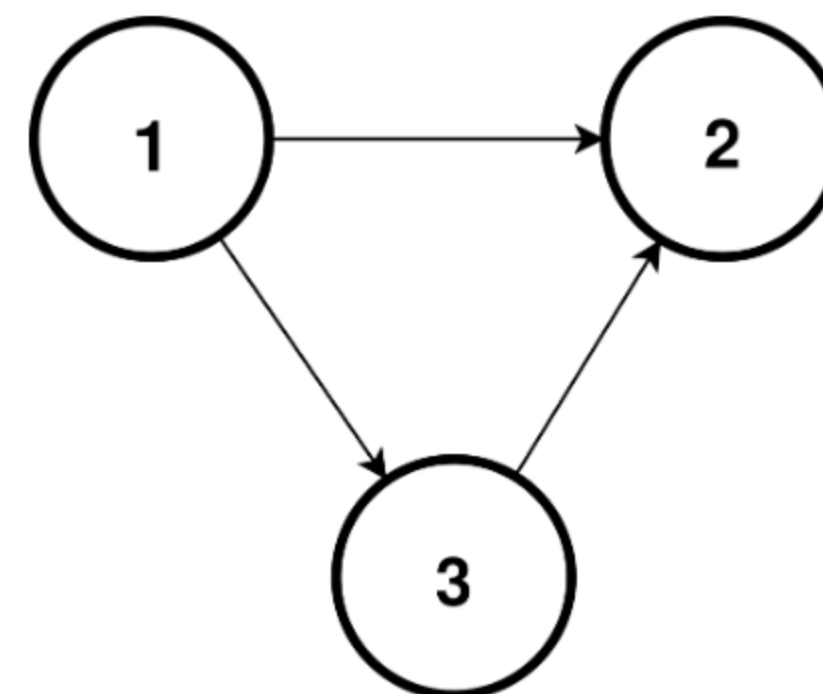
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- Shuffling allows us to convert between the orderings



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How do we do it?

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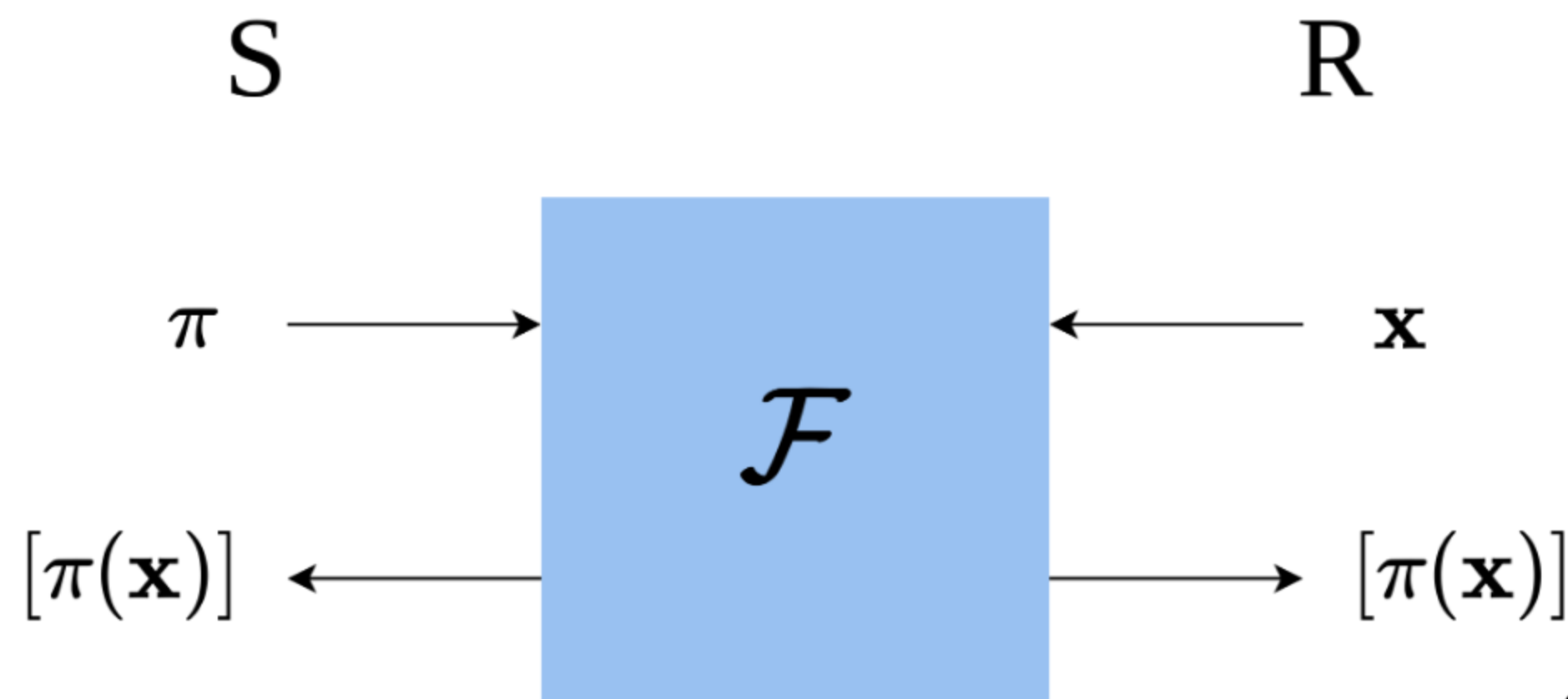
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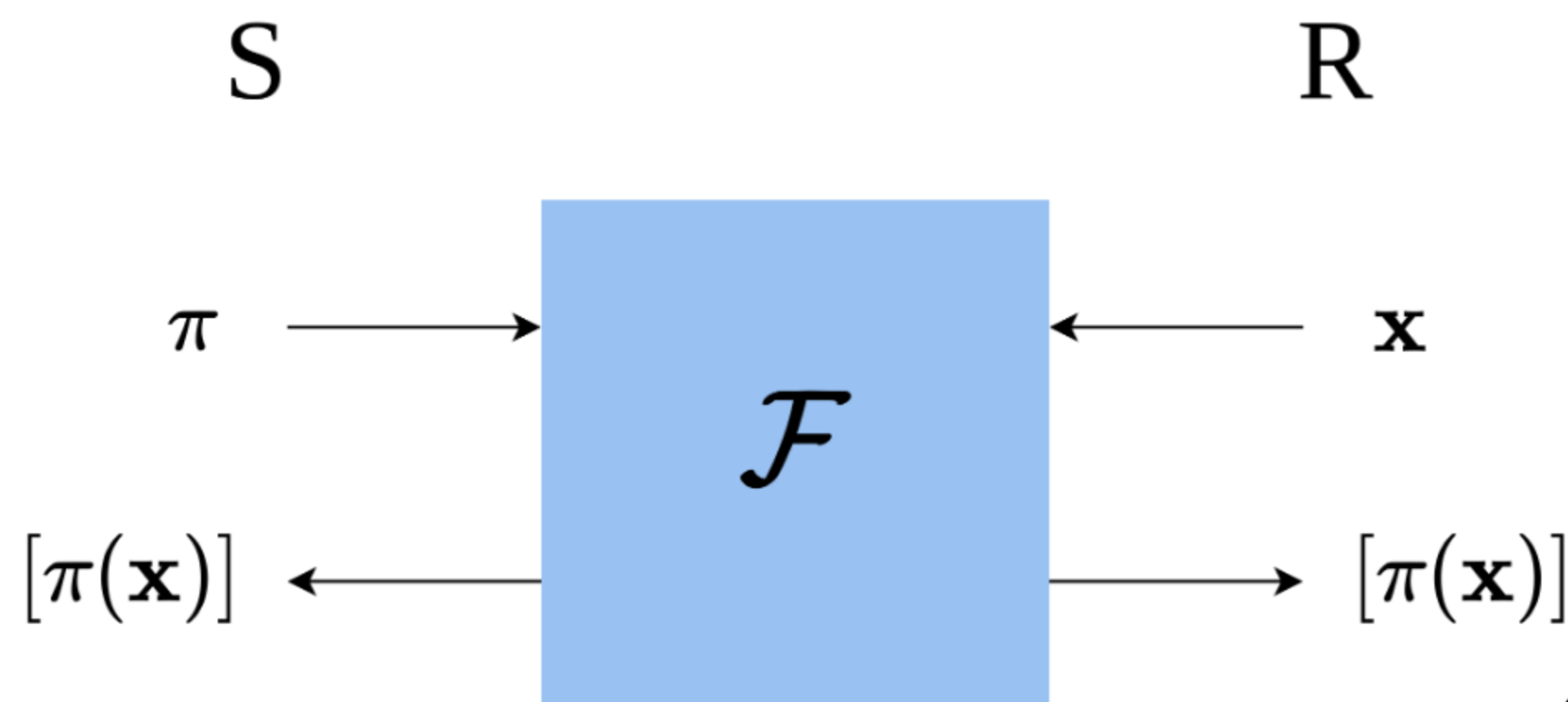
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- Known as "Permute+Share"



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$$\pi(A) = B + C$$

Online Derandomization Protocol

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S: (π, C)

R: $(\mathbf{x}, (A, B))$

$\mathbf{x} - A$



$\pi(\mathbf{x} - A) + C$

B

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$\xleftarrow{\mathbf{x} - A}$

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$$\begin{aligned}\text{output} &= \pi(\mathbf{x} - A) + C + B \\ &= \pi(\mathbf{x} - A) + \pi(A) \\ &= \pi(\mathbf{x}) - \pi(A) + \pi(A) \\ &= \pi(\mathbf{x})\end{aligned}$$

Formally Defining the Problem

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Functionality $\mathcal{F}_{\text{GenPerm}}$

PUBLIC PARAMETERS: Number of elements n and bitwidth ℓ .

INPUT: None

OUTPUT: The functionality samples a uniformly random permutation $\pi \leftarrow S_n$ and uniformly random vectors $A, B, C \in \{0, 1\}^{n \times \ell}$, such that $\pi(A) = B + C$. $\mathcal{F}_{\text{GenPerm}}$ outputs (π, C) to the sender and (A, B) to the receiver.

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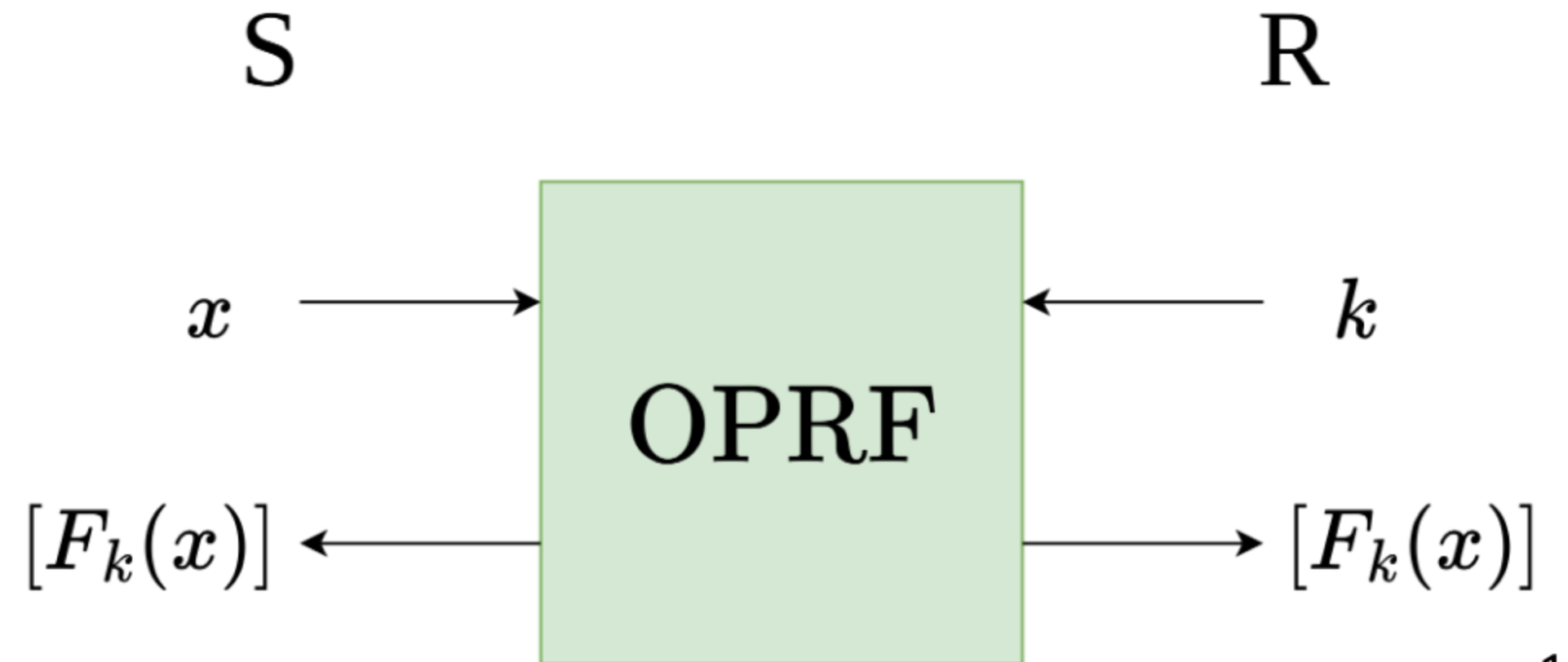
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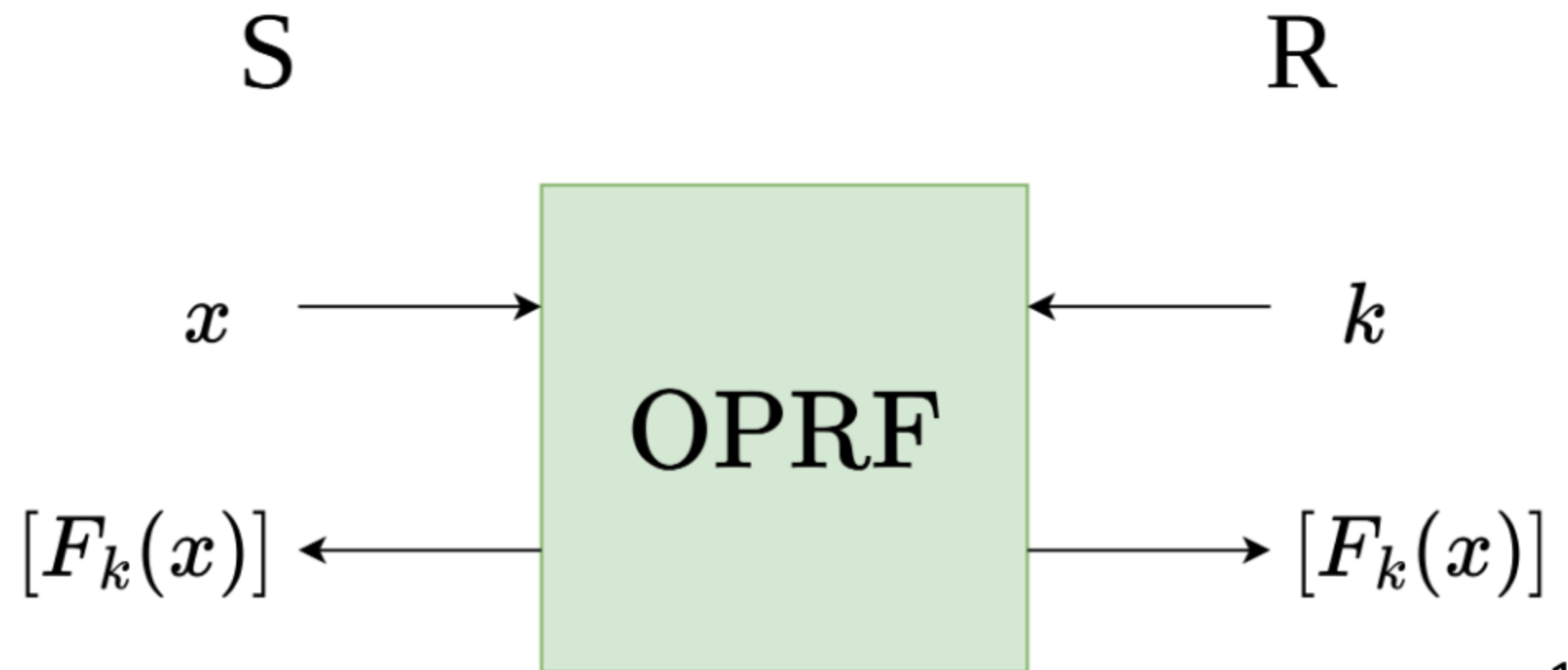
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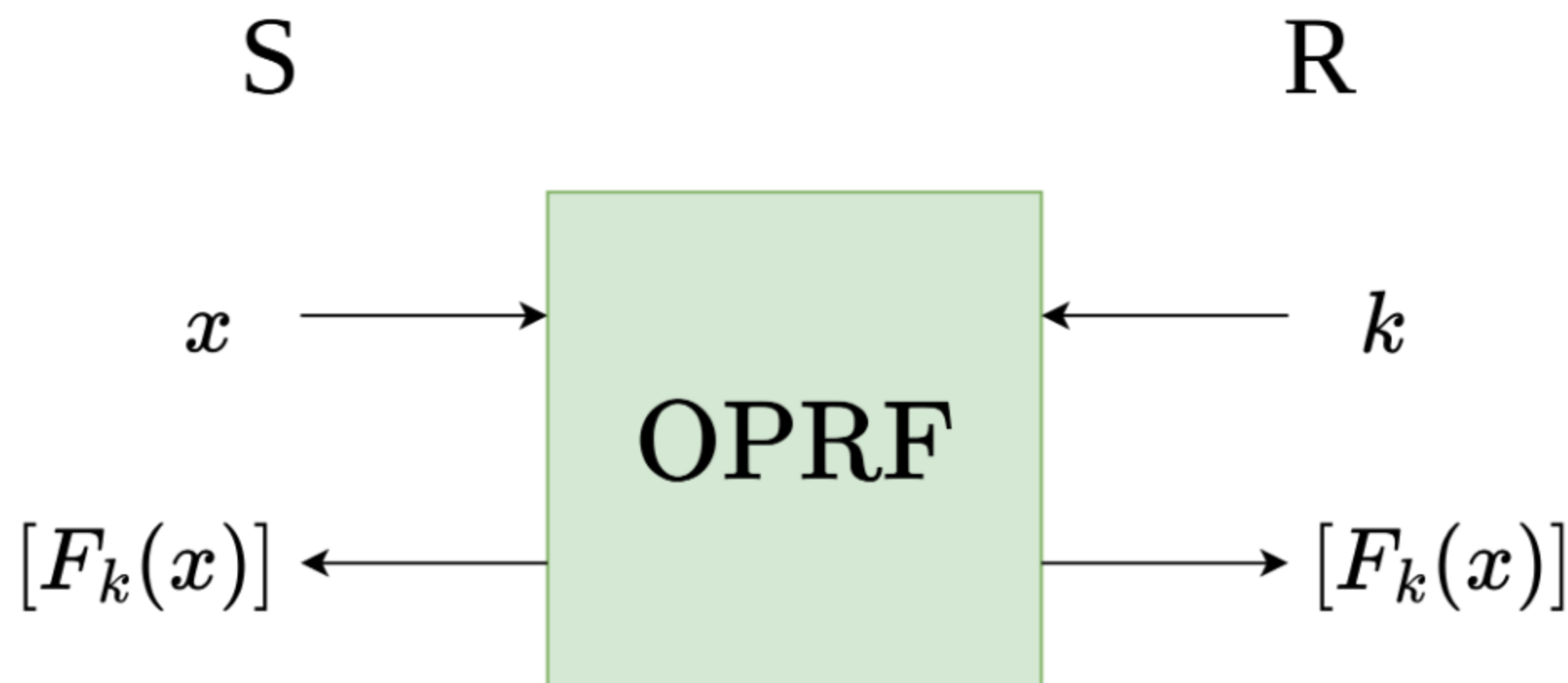
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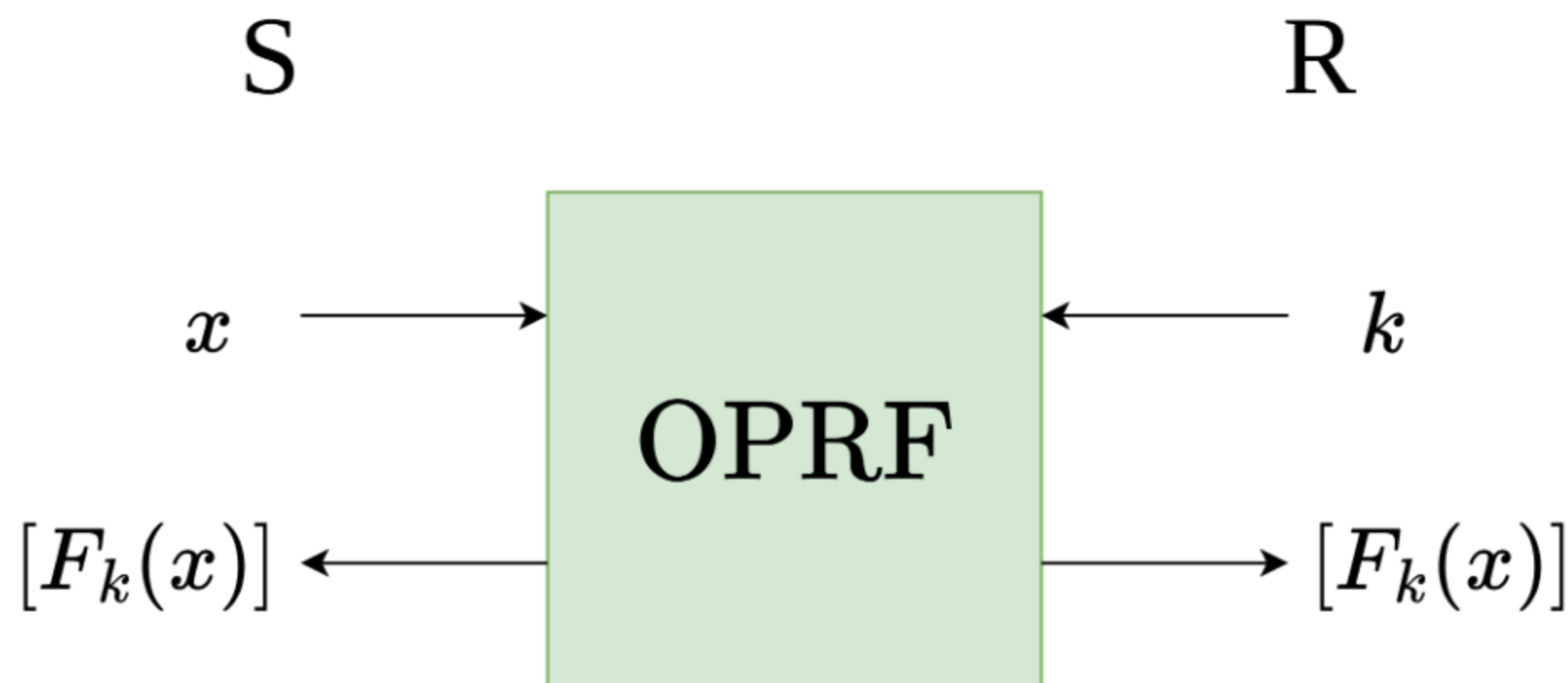
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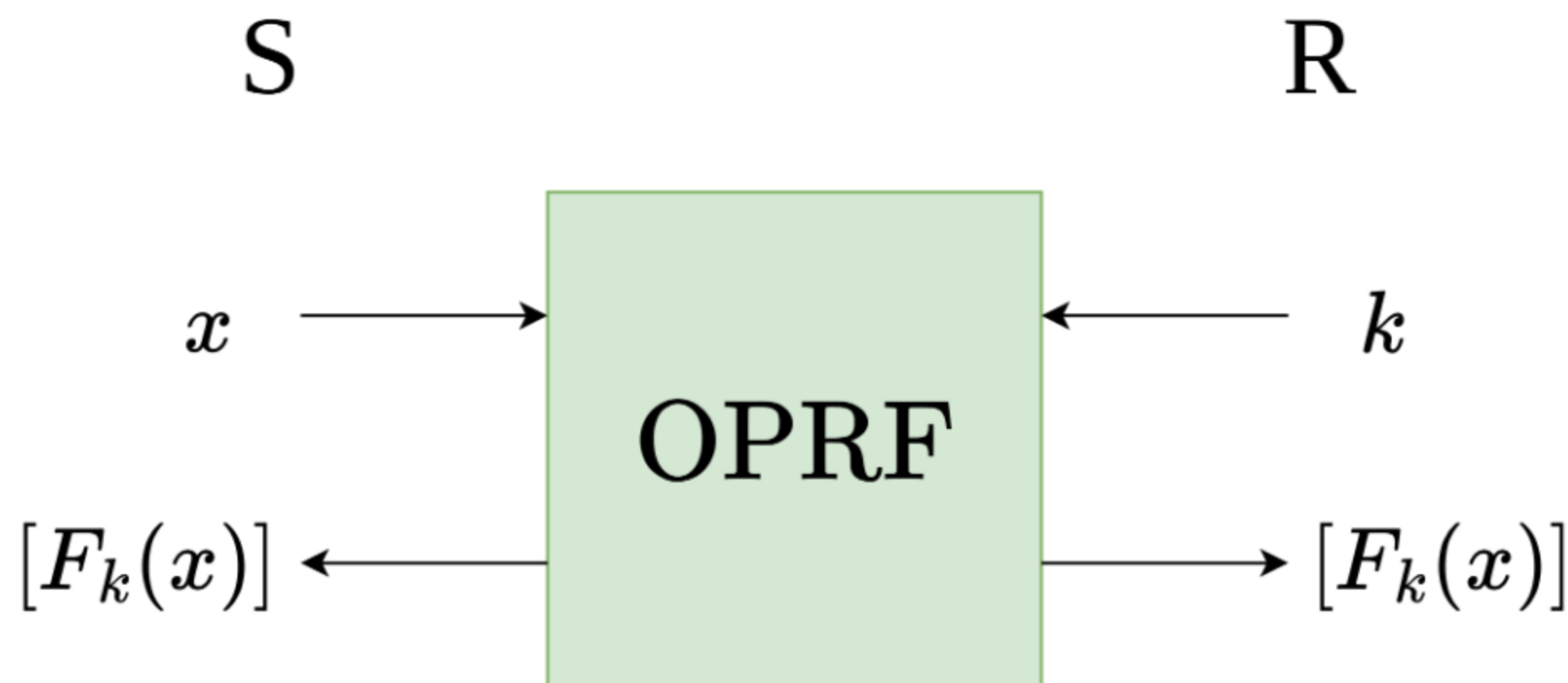
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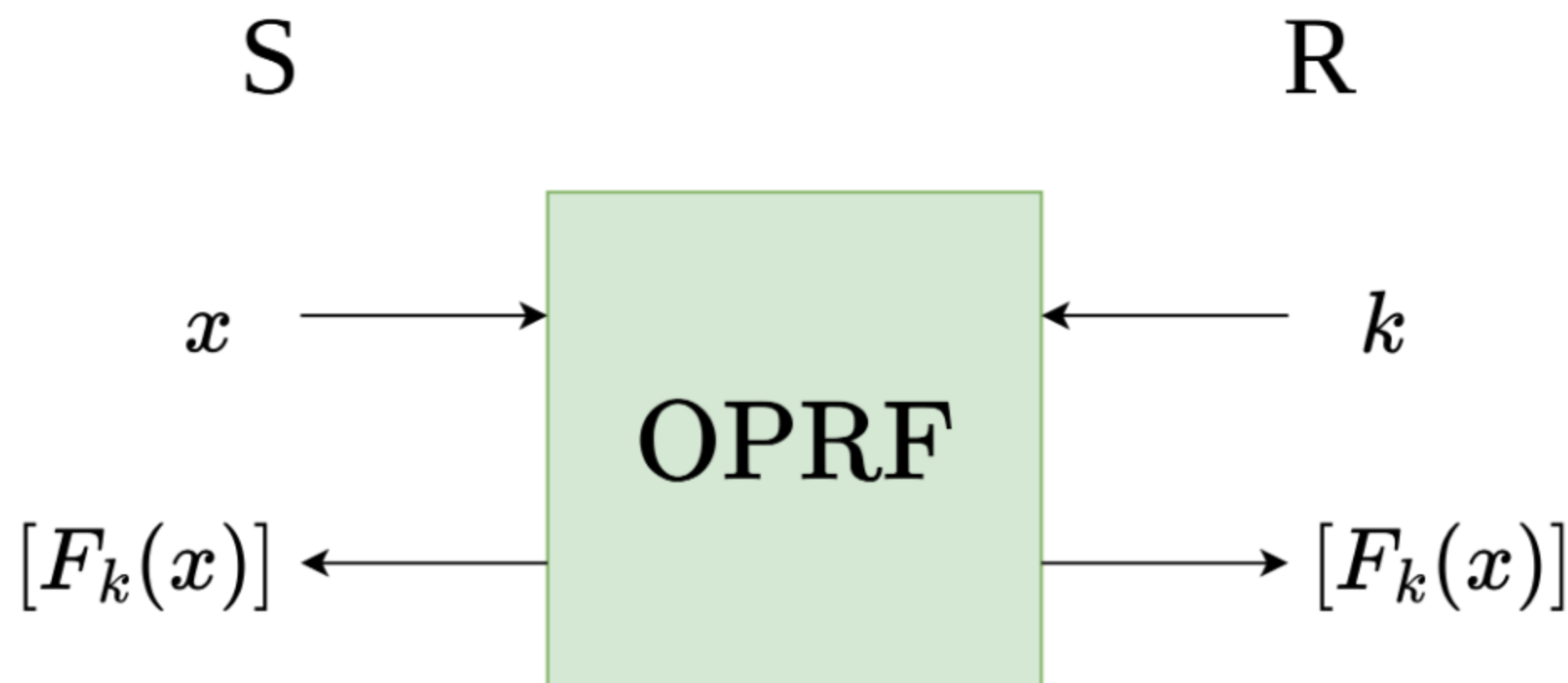
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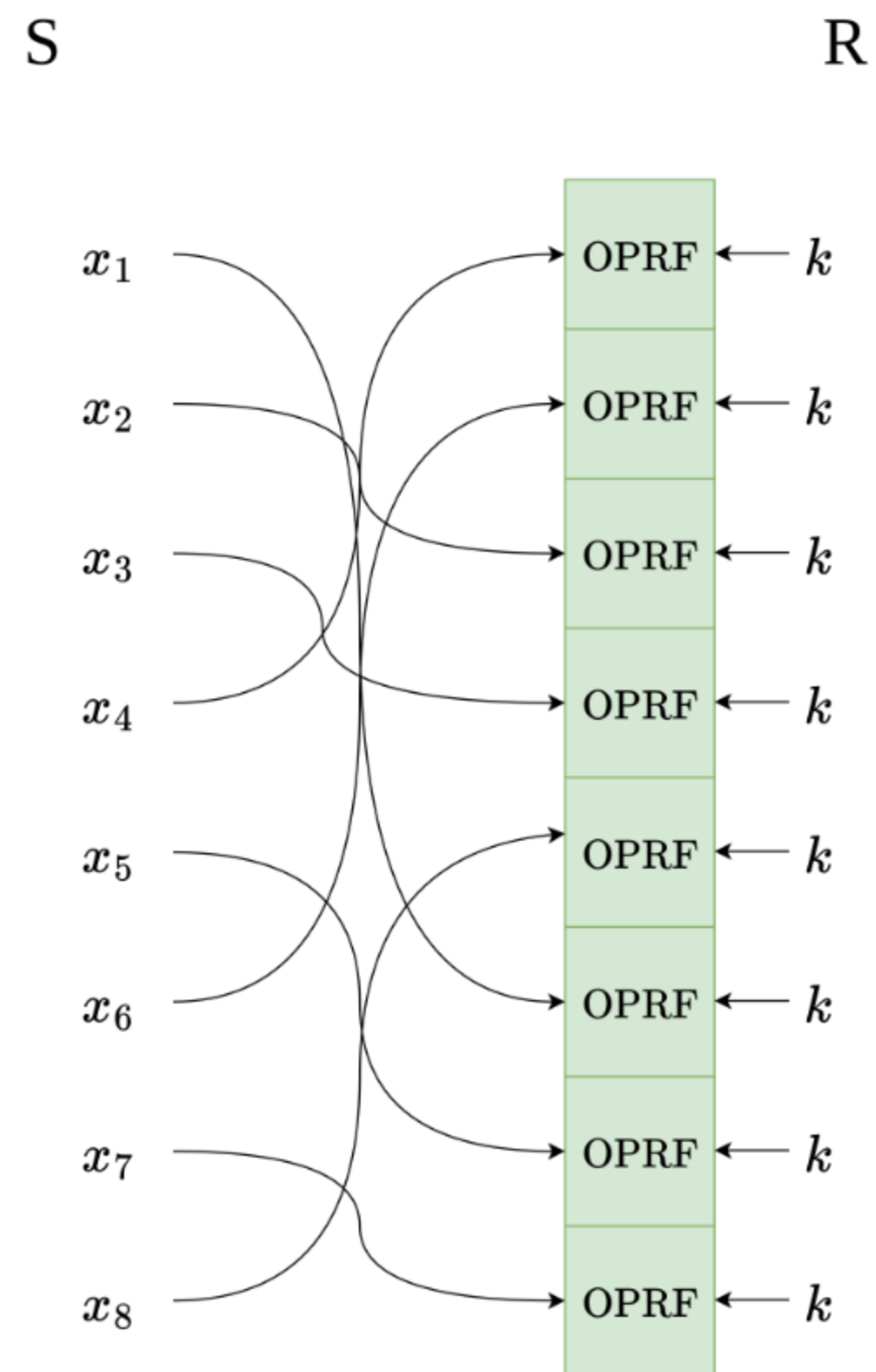
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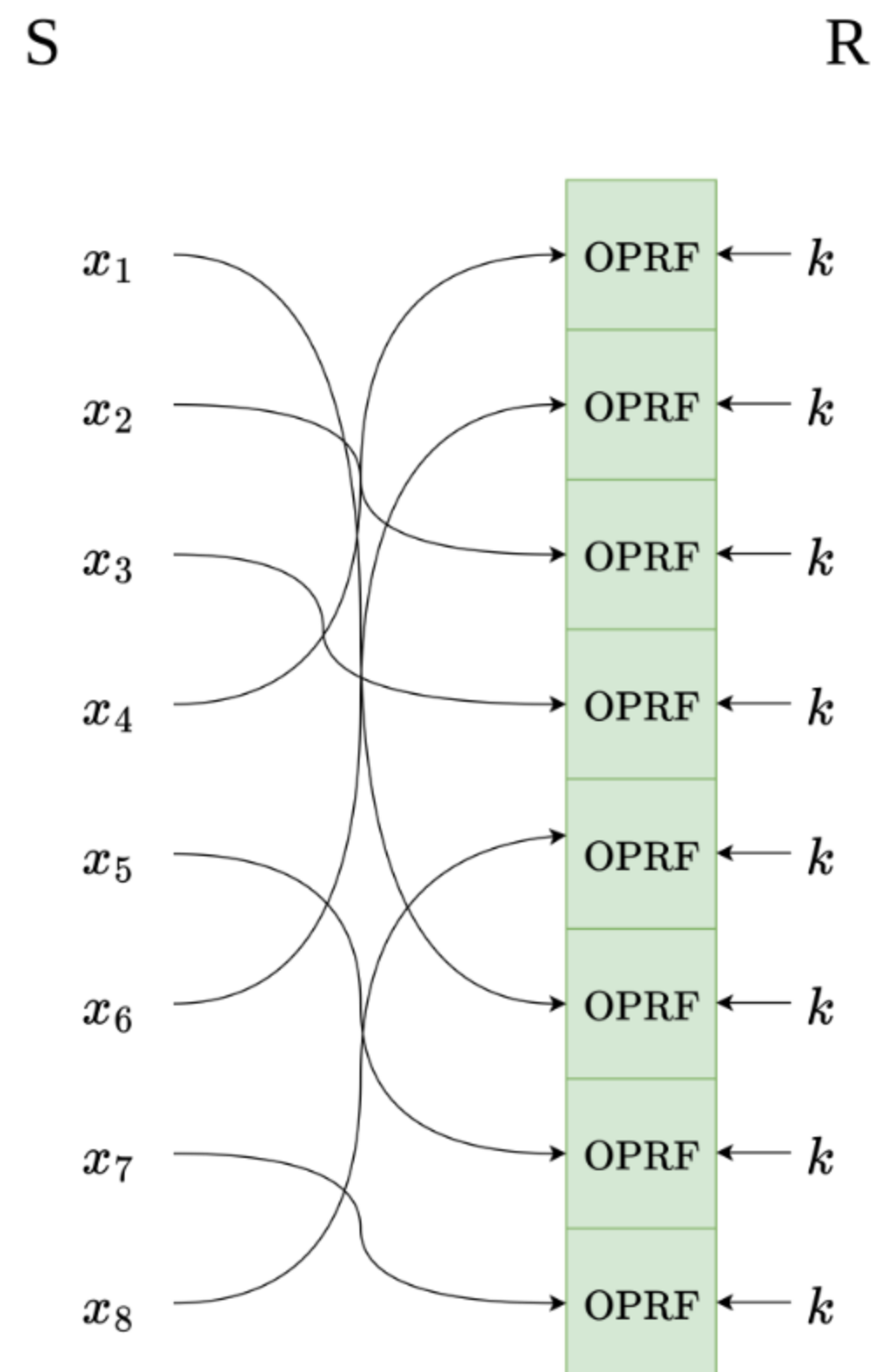


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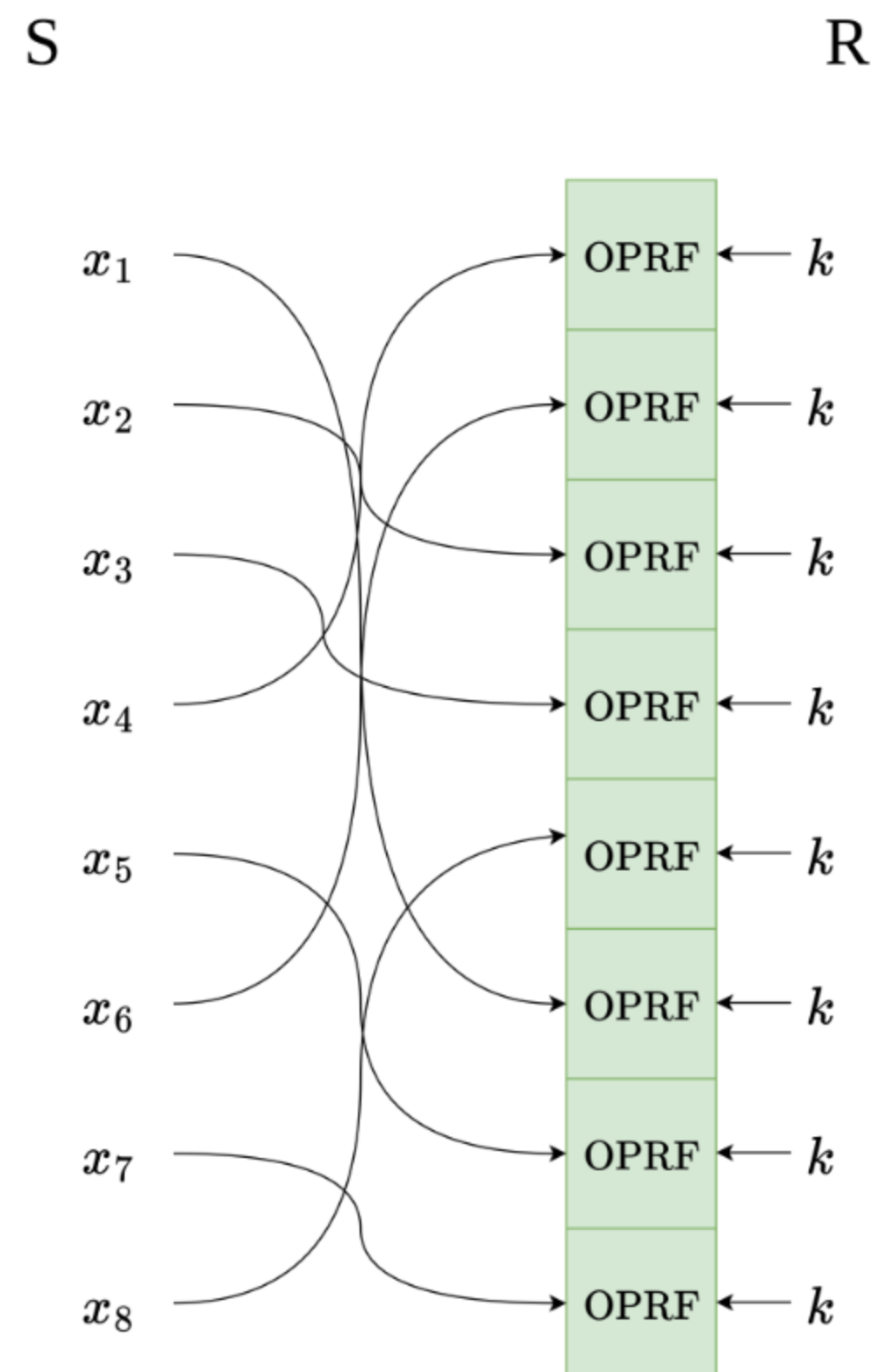
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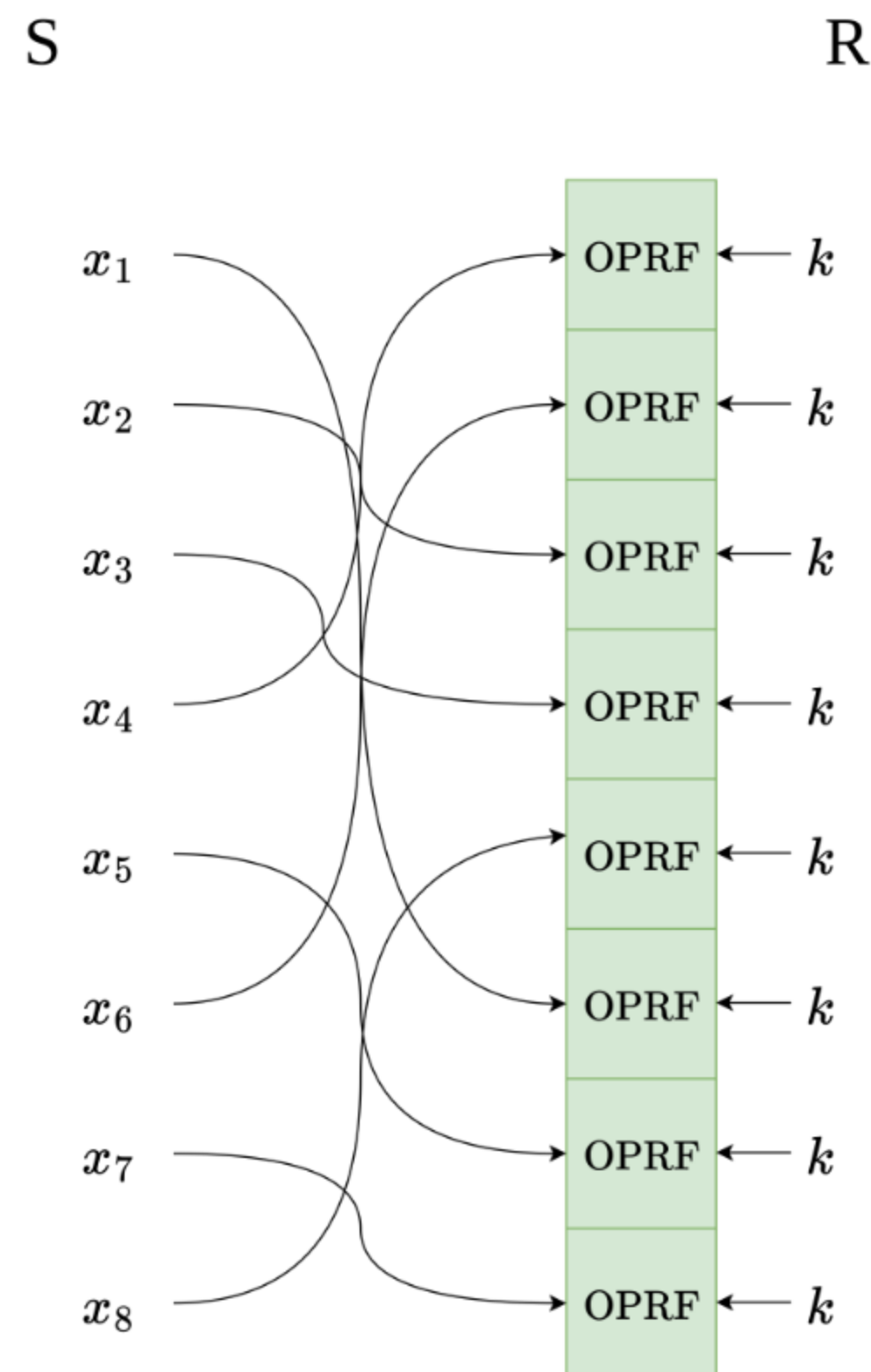
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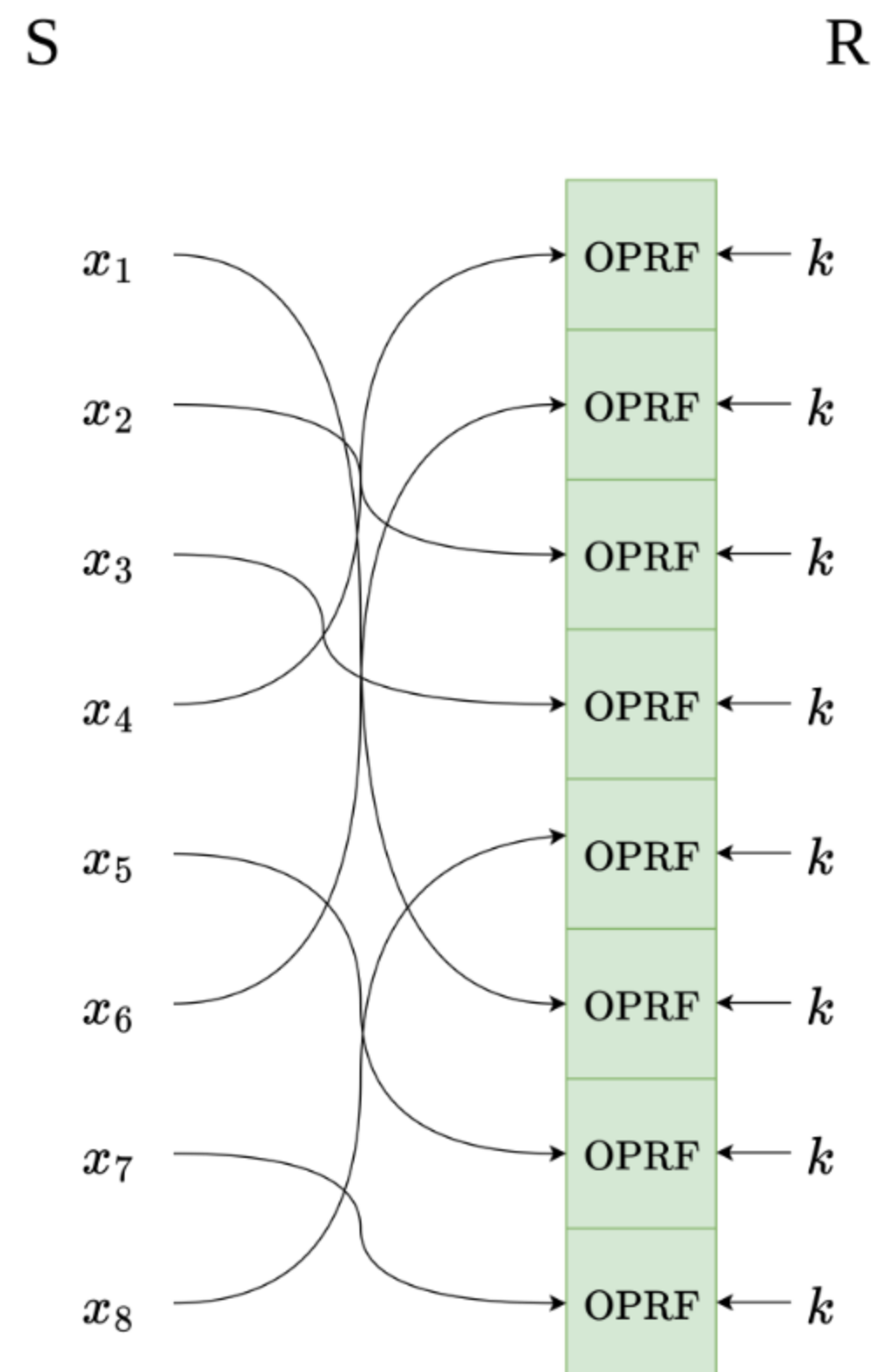
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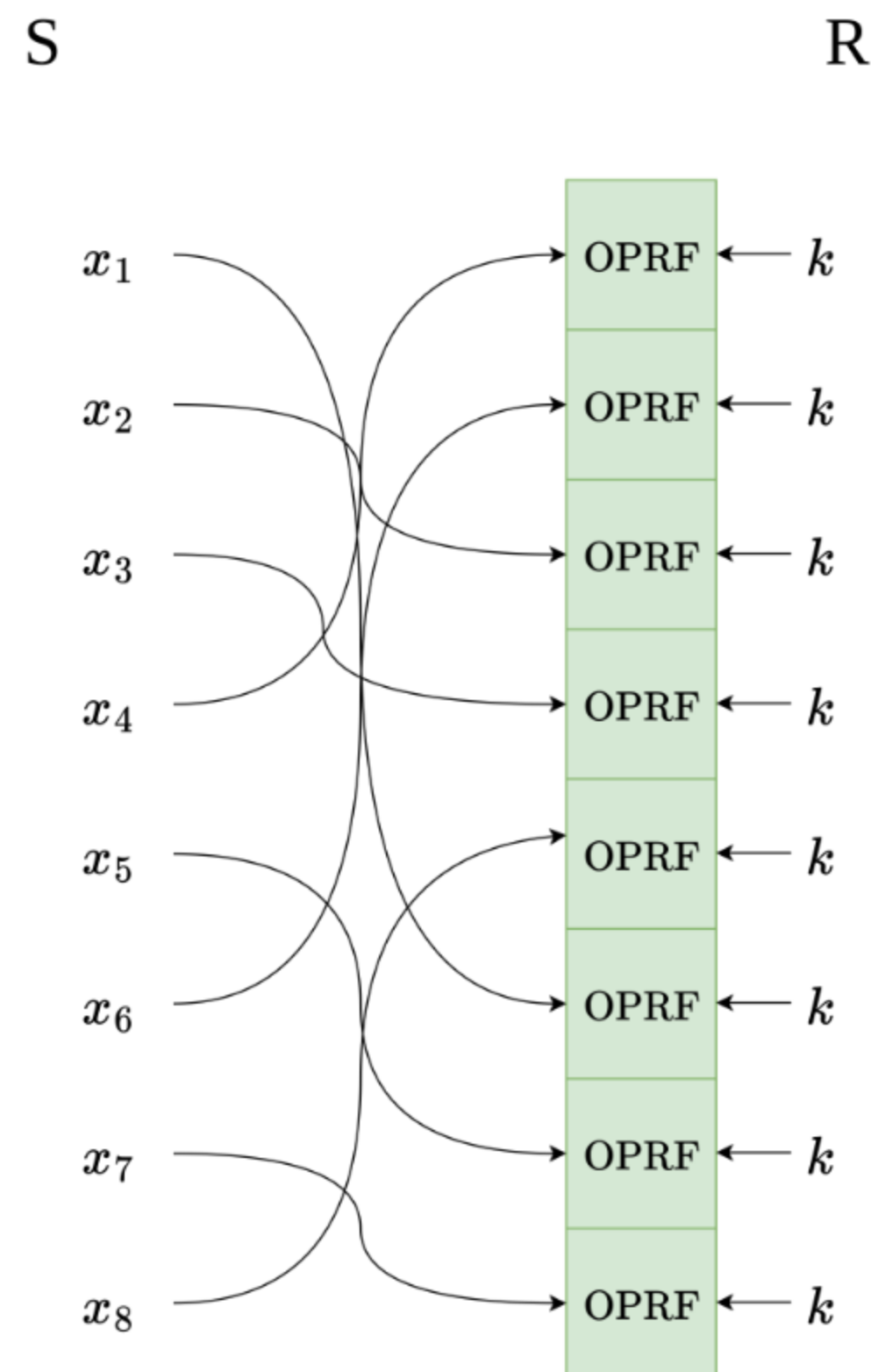
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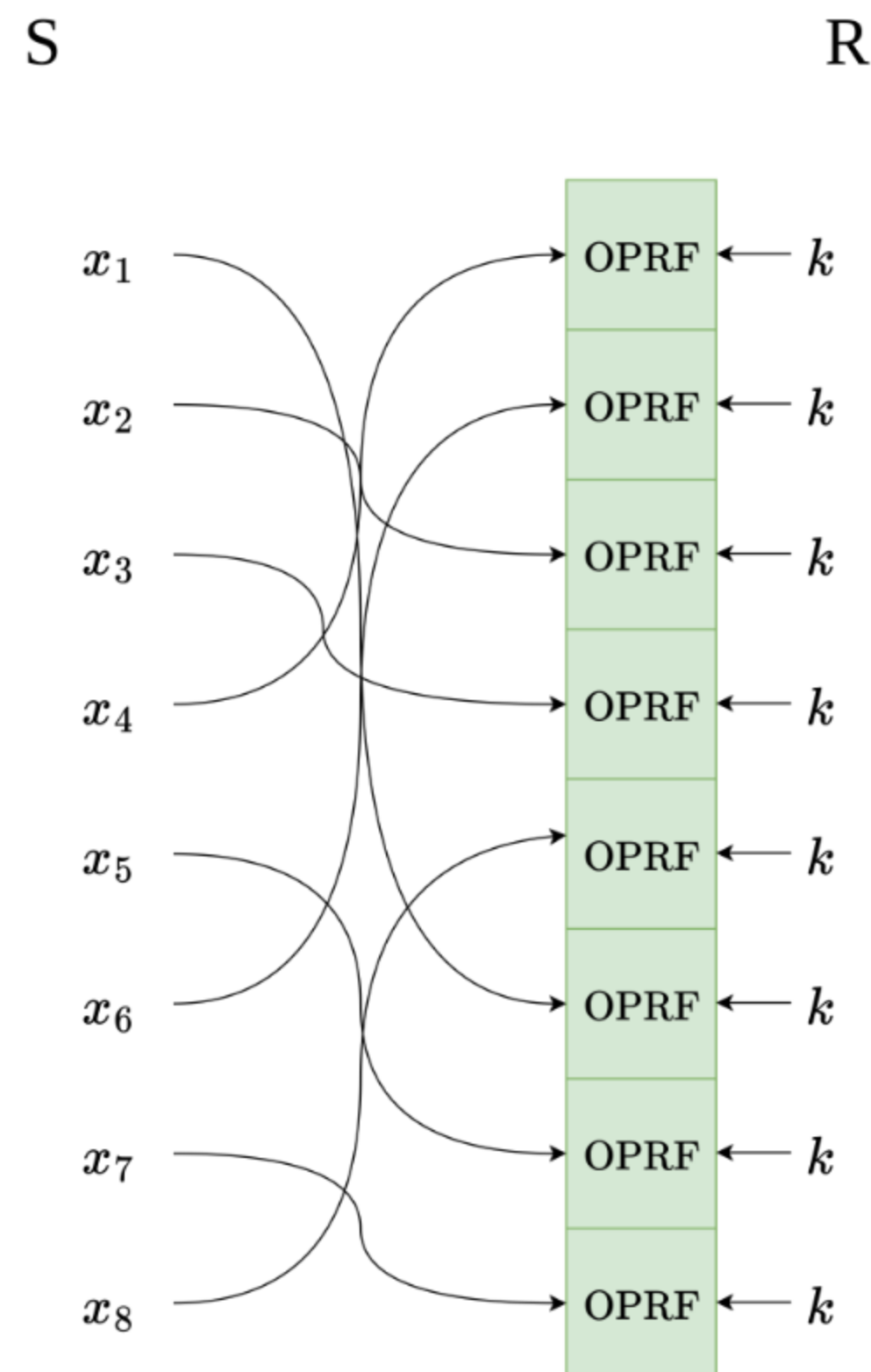
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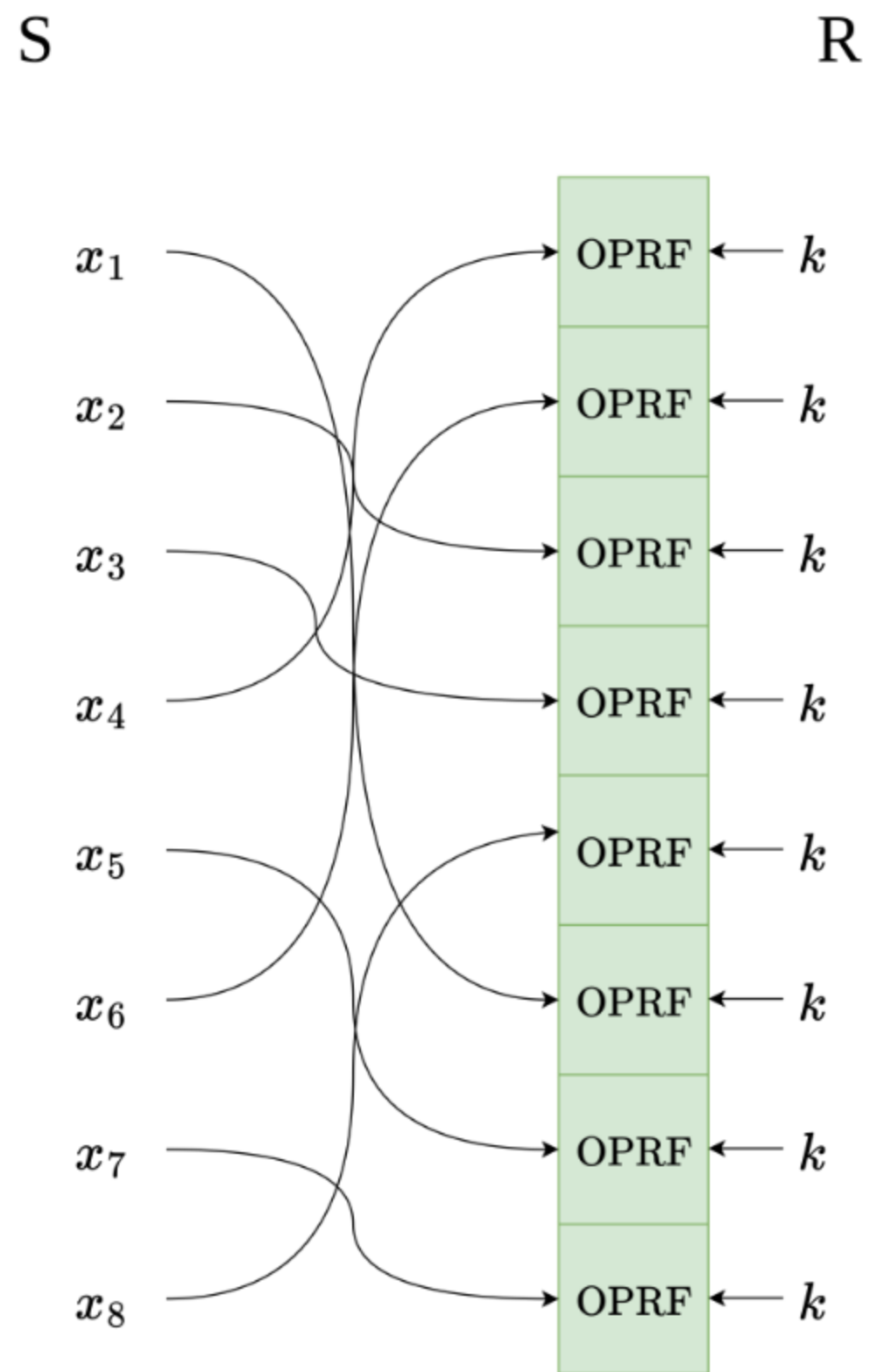


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- Parties set B and C as the OPRF outputs

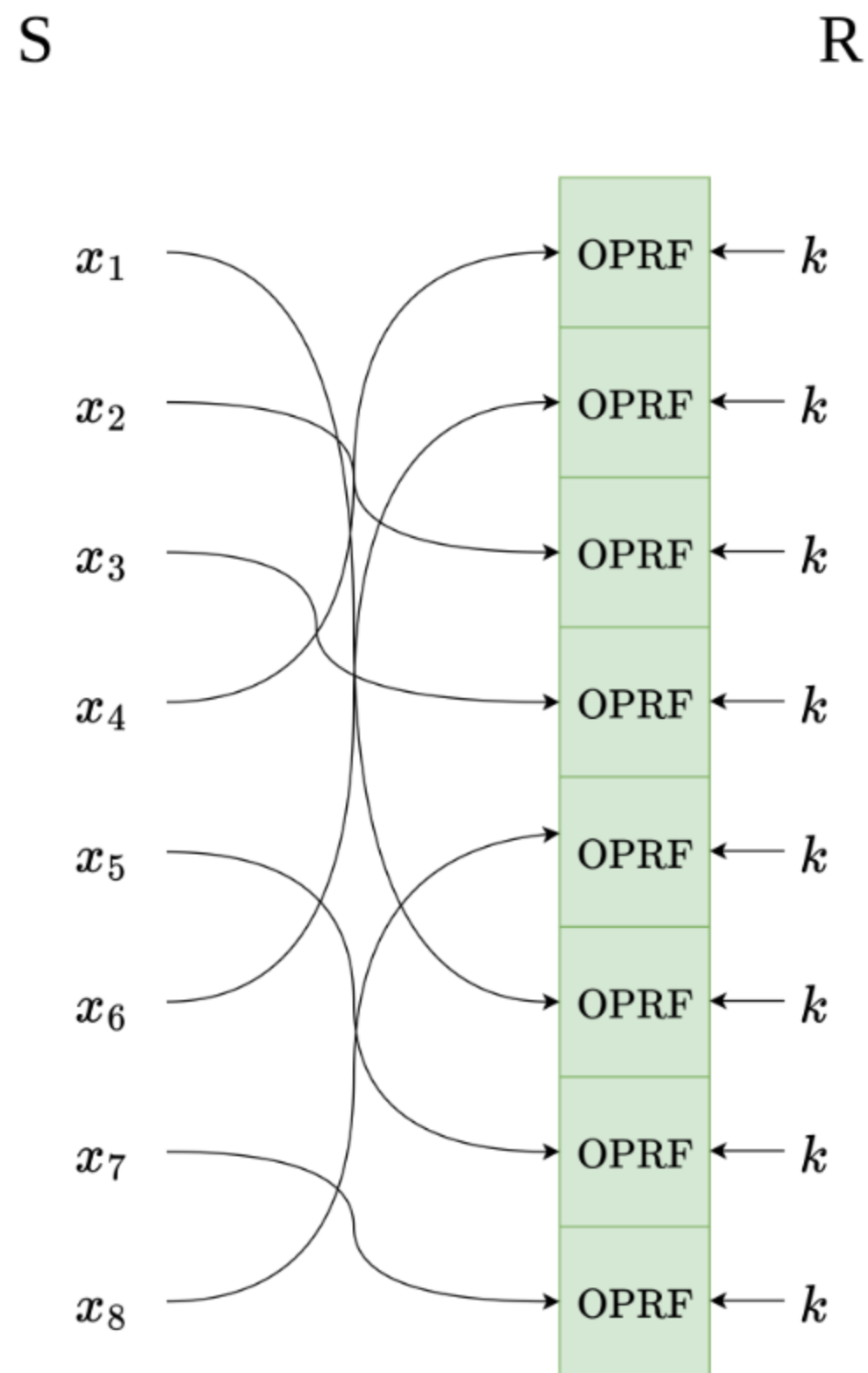


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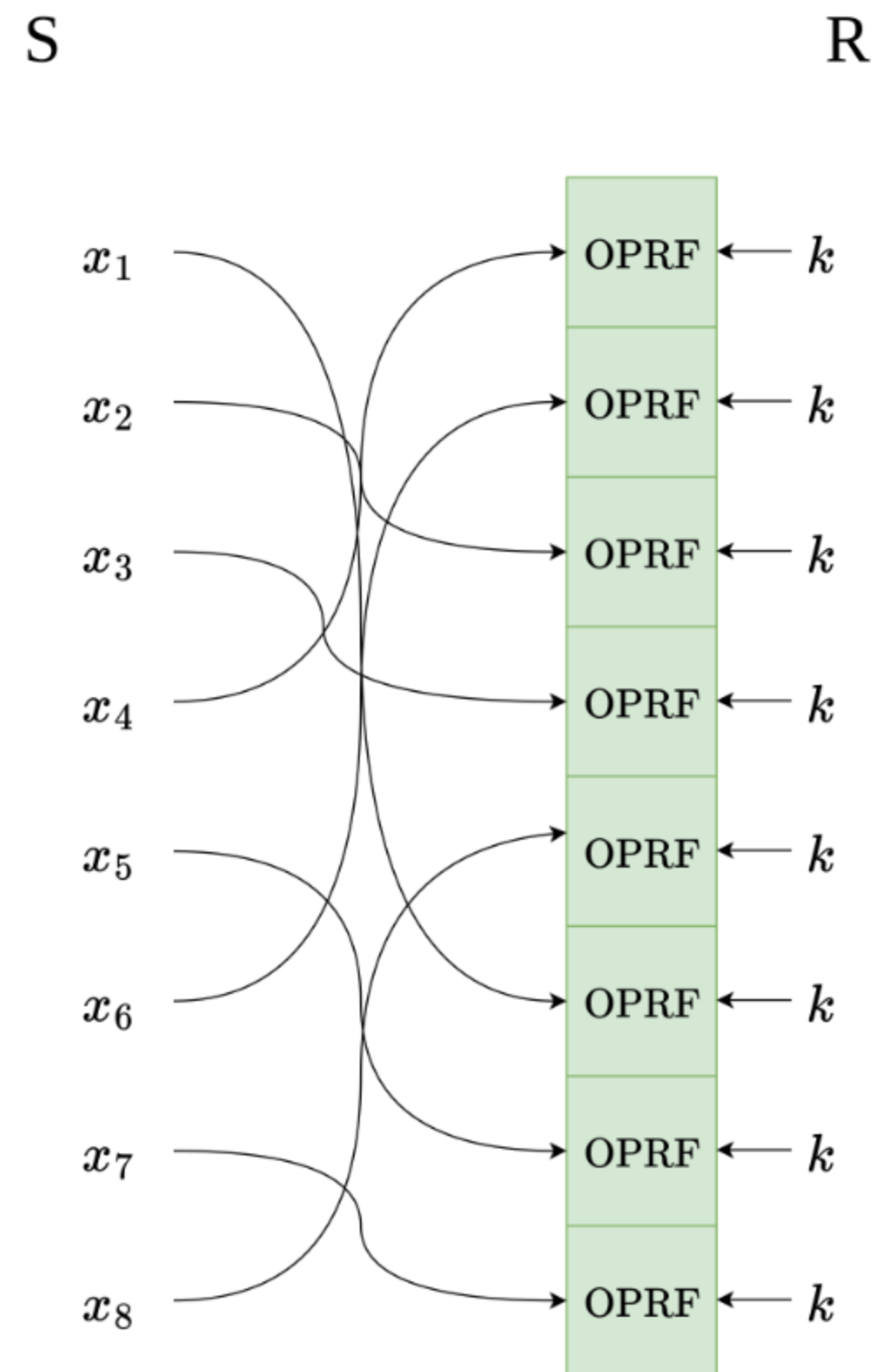
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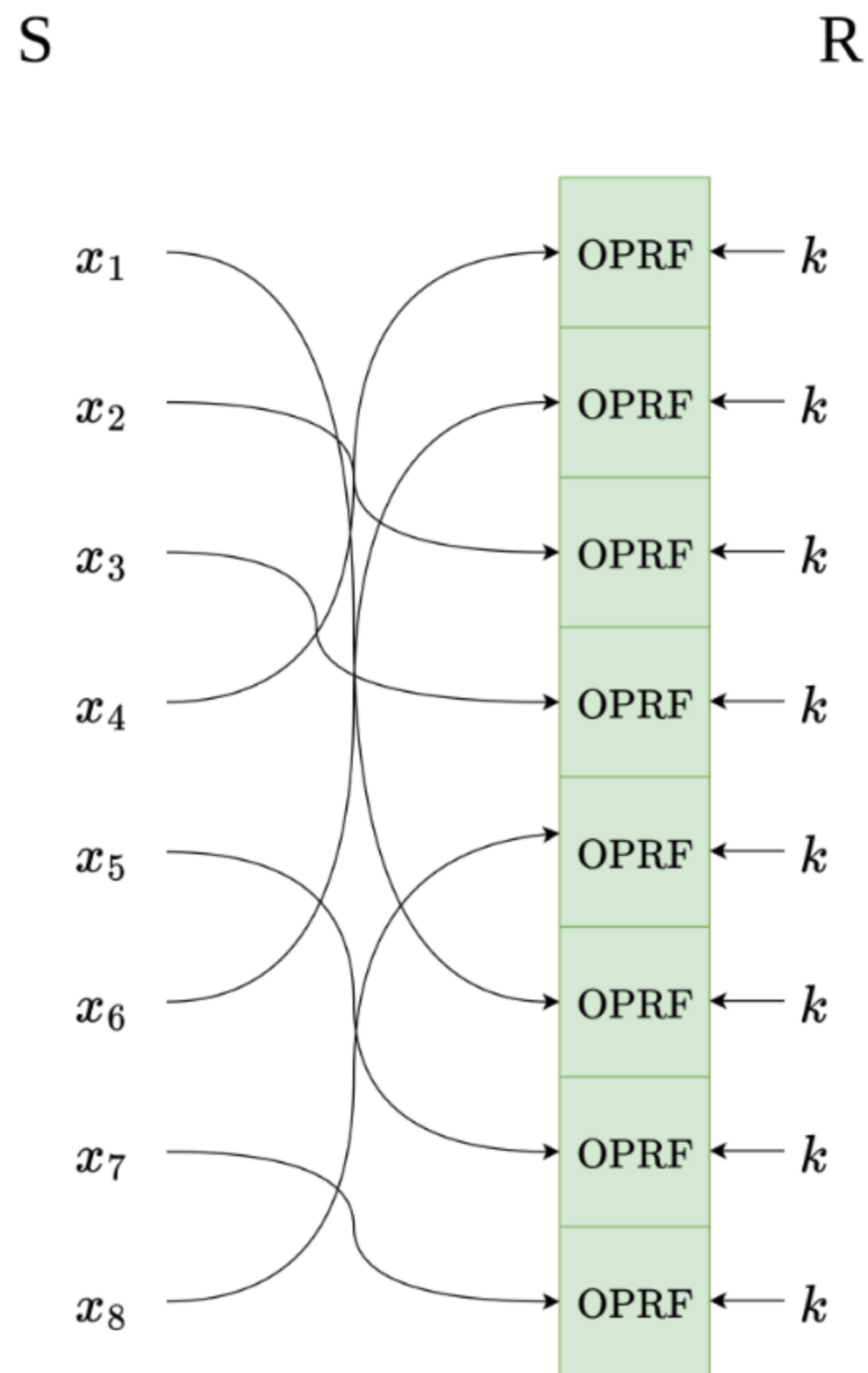
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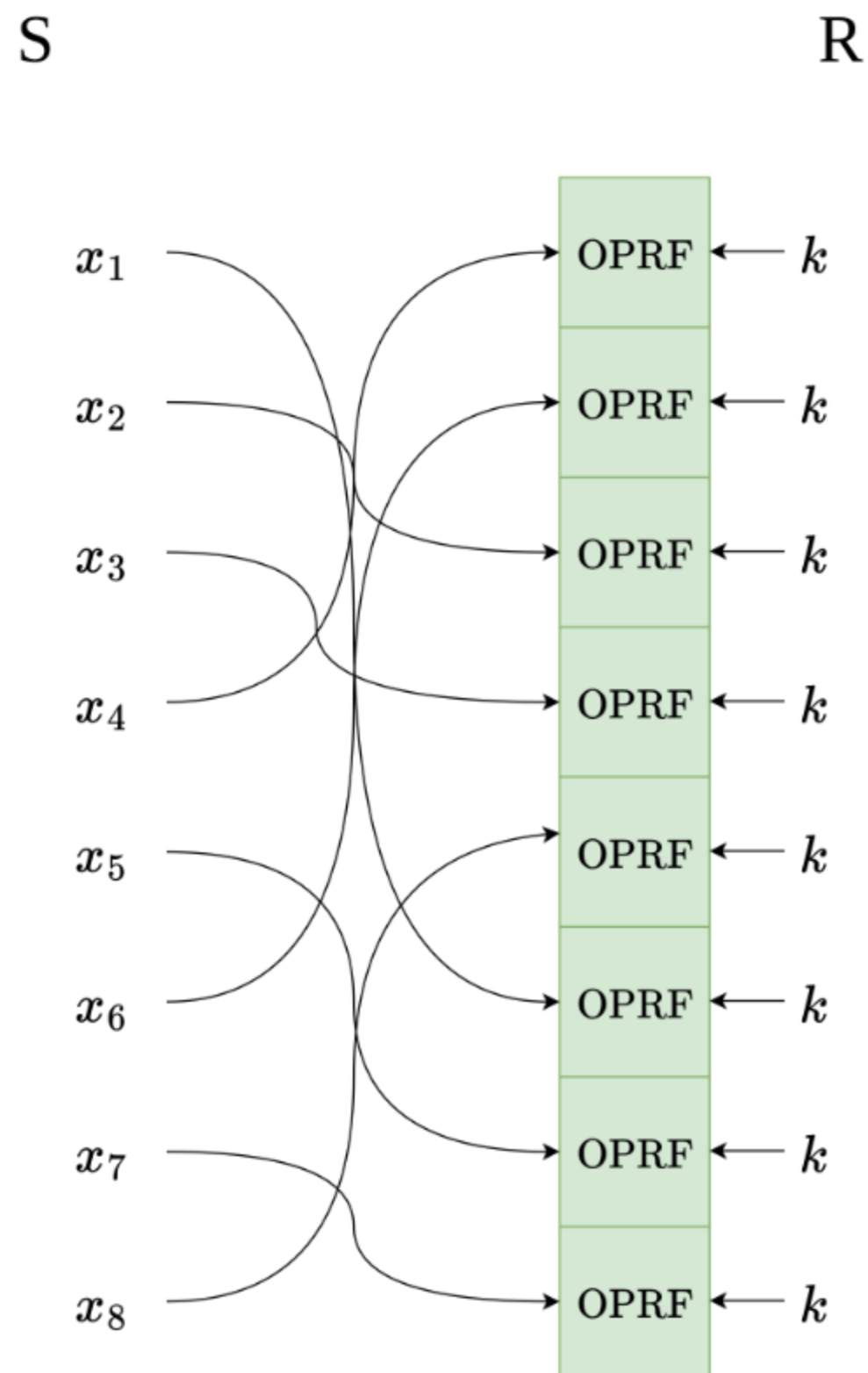
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- $\approx 4s$ per million elements



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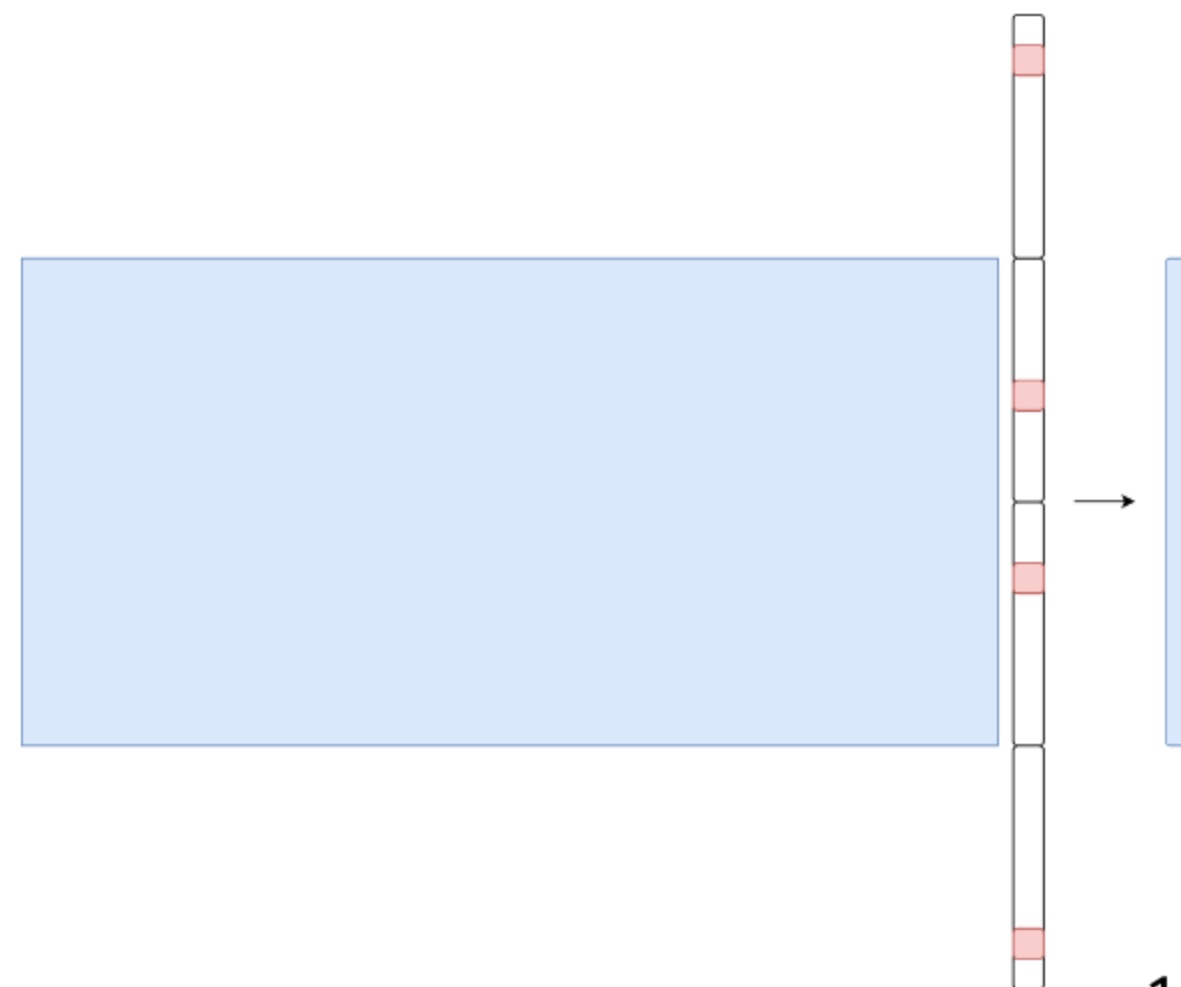
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Scaling with Many Parties

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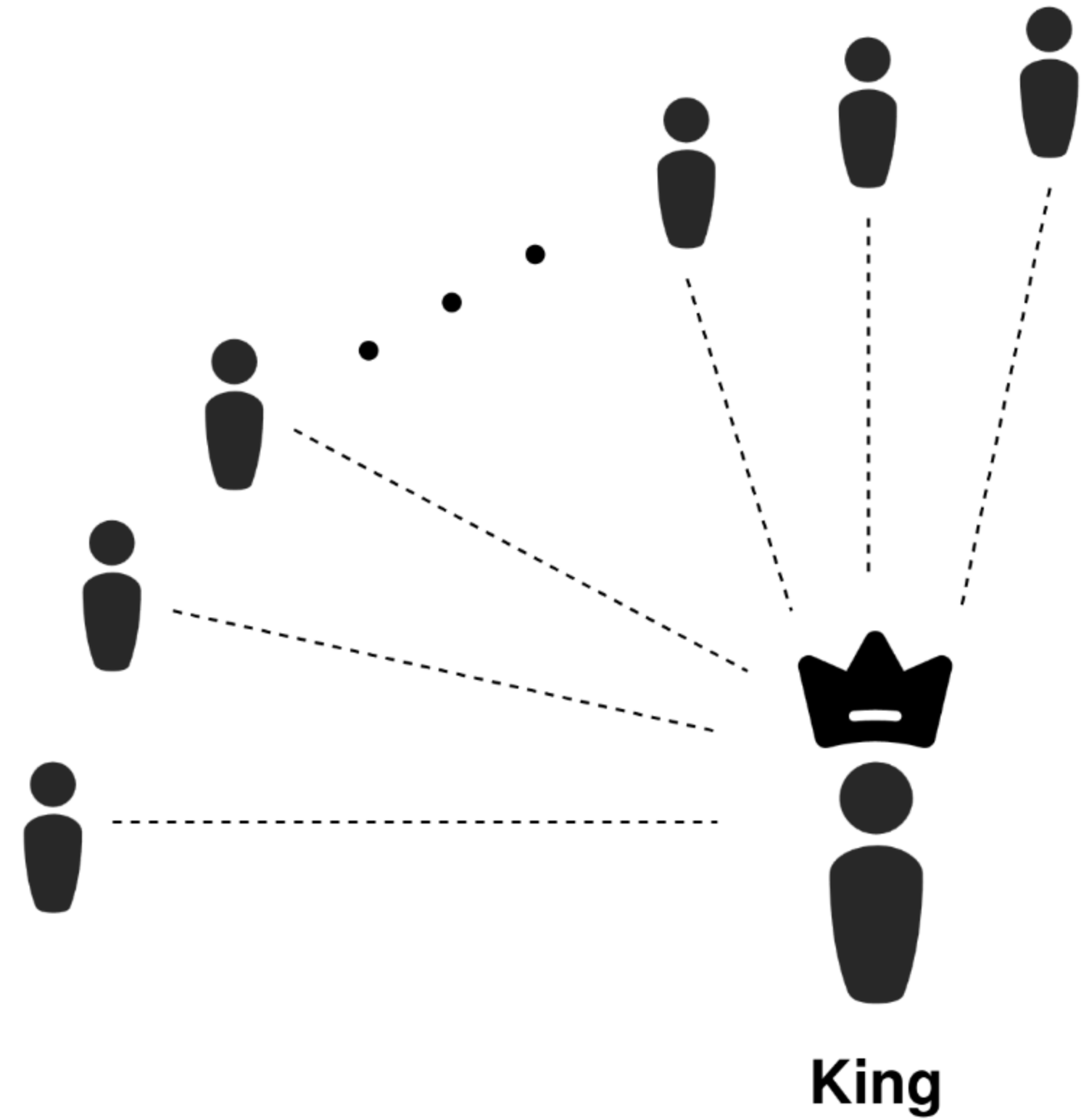
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Goal: Scale *practical* MPC to thousands of parties.

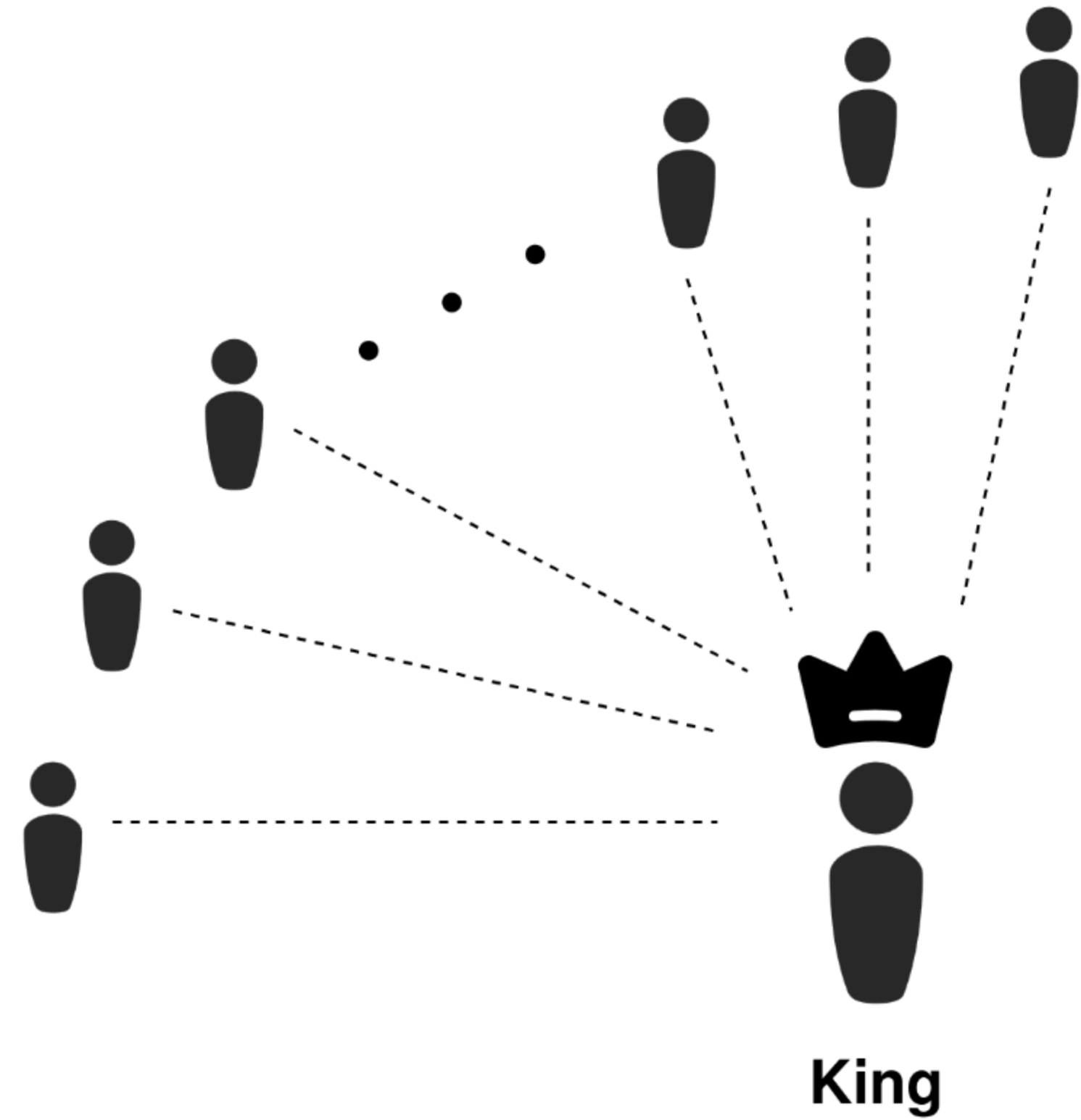
A Star Topology

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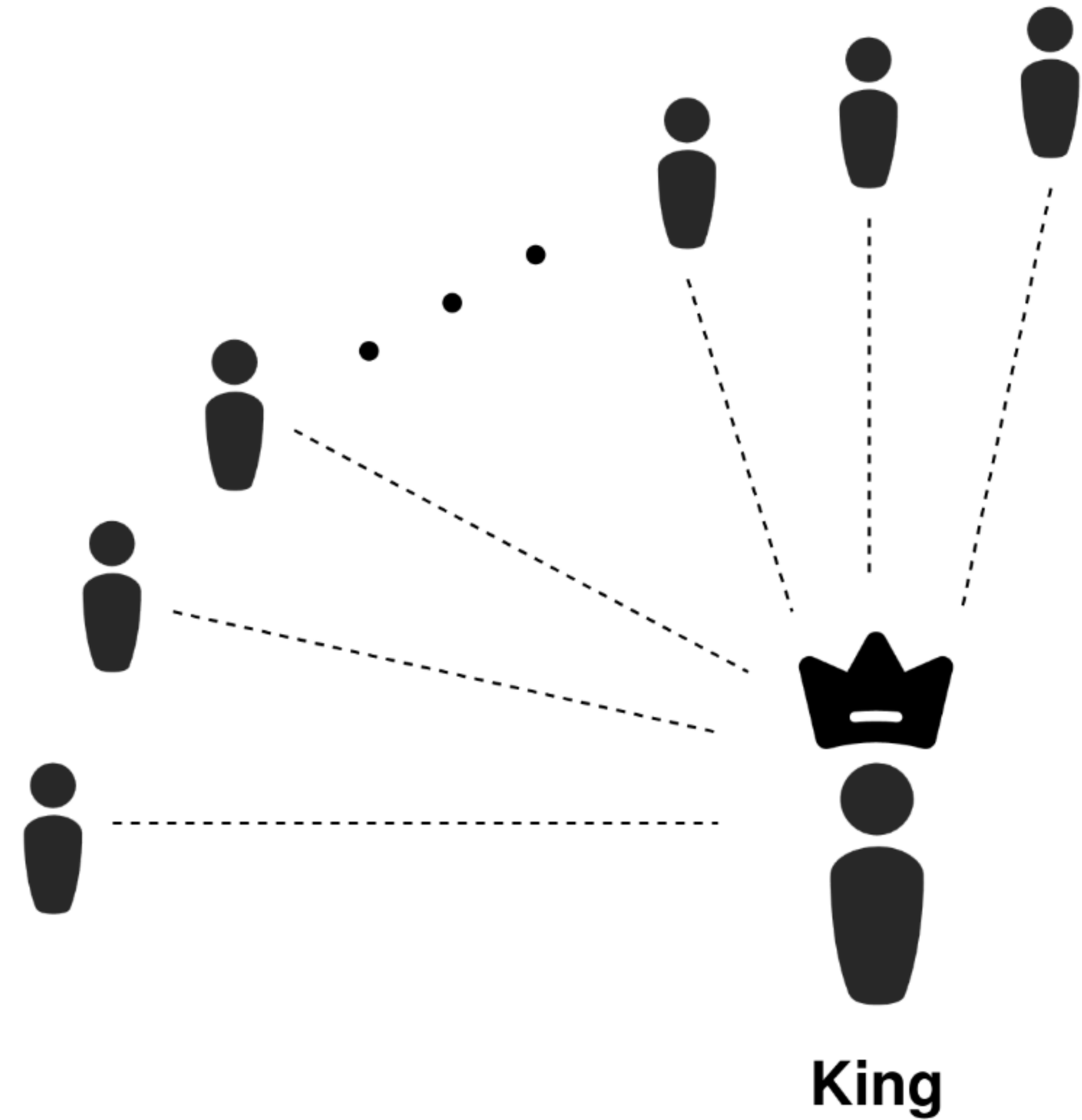
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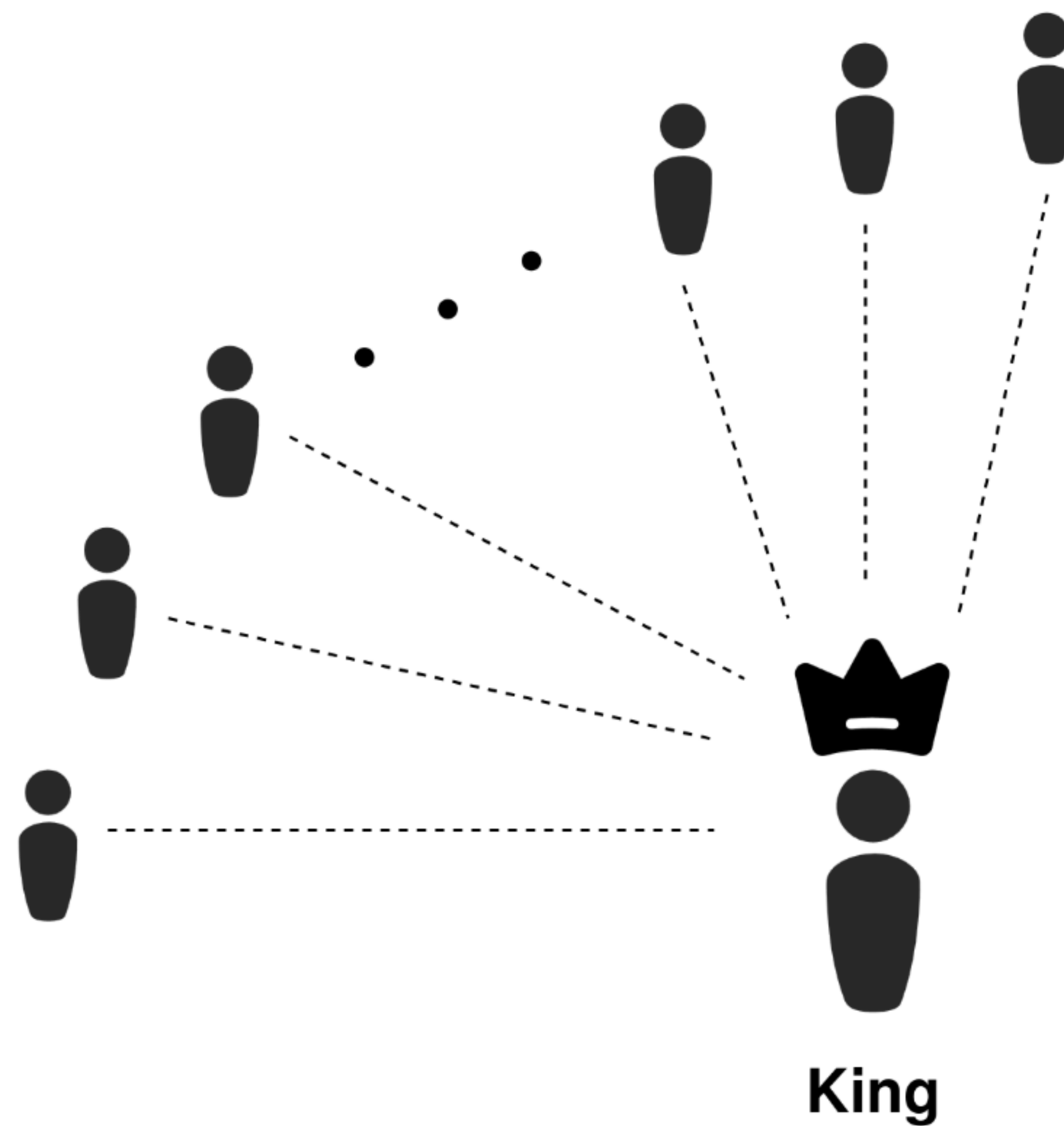
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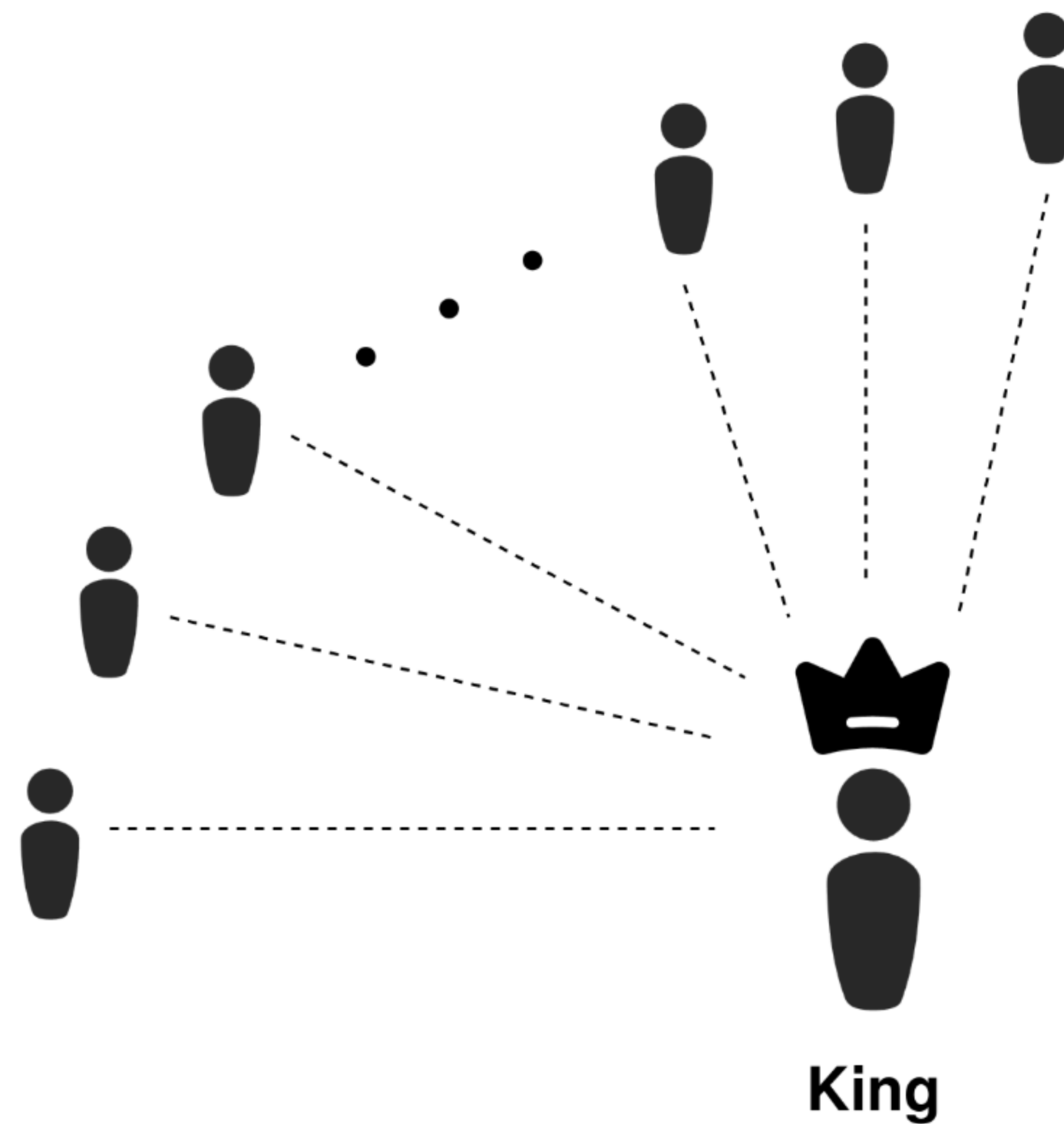
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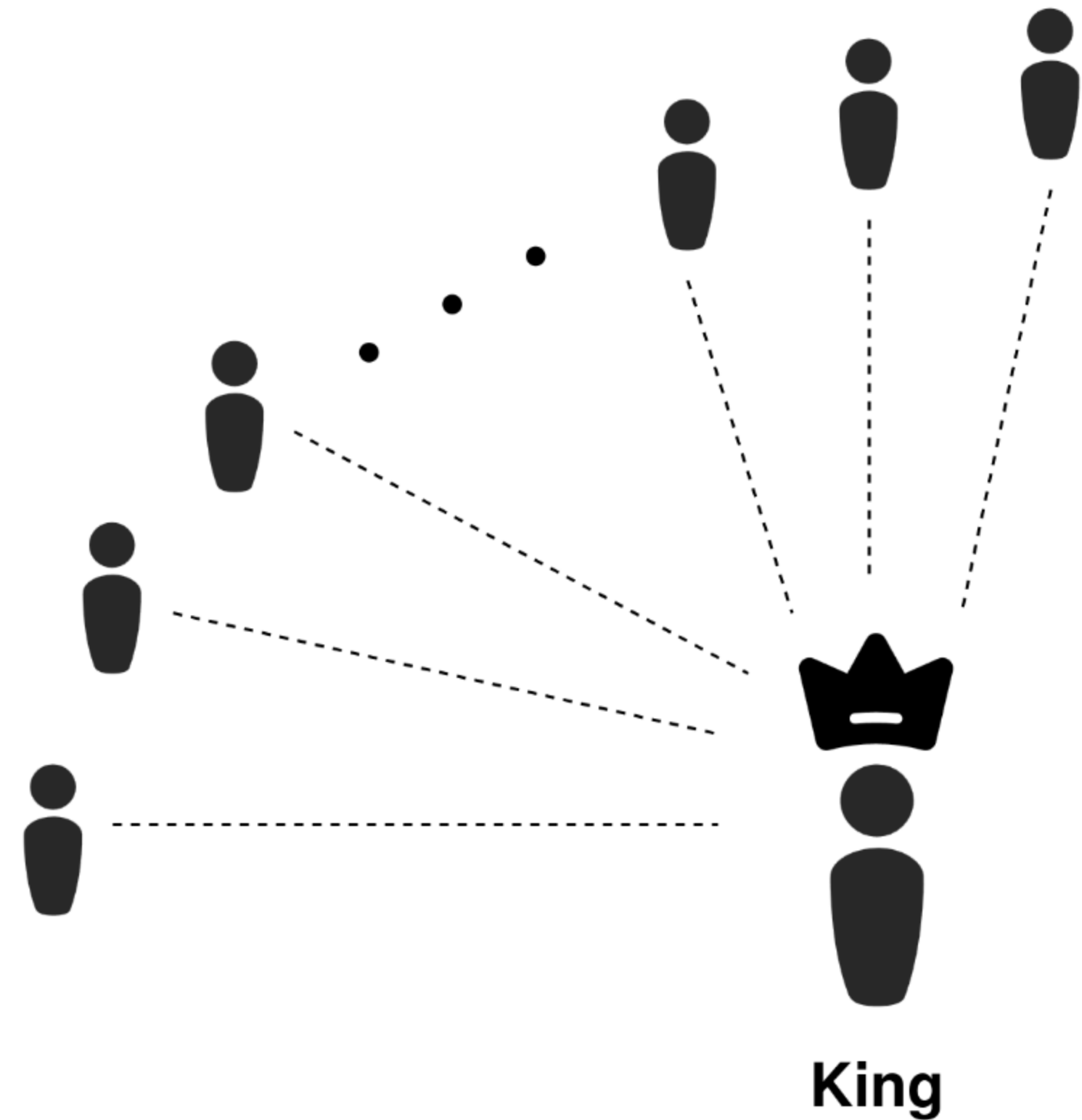
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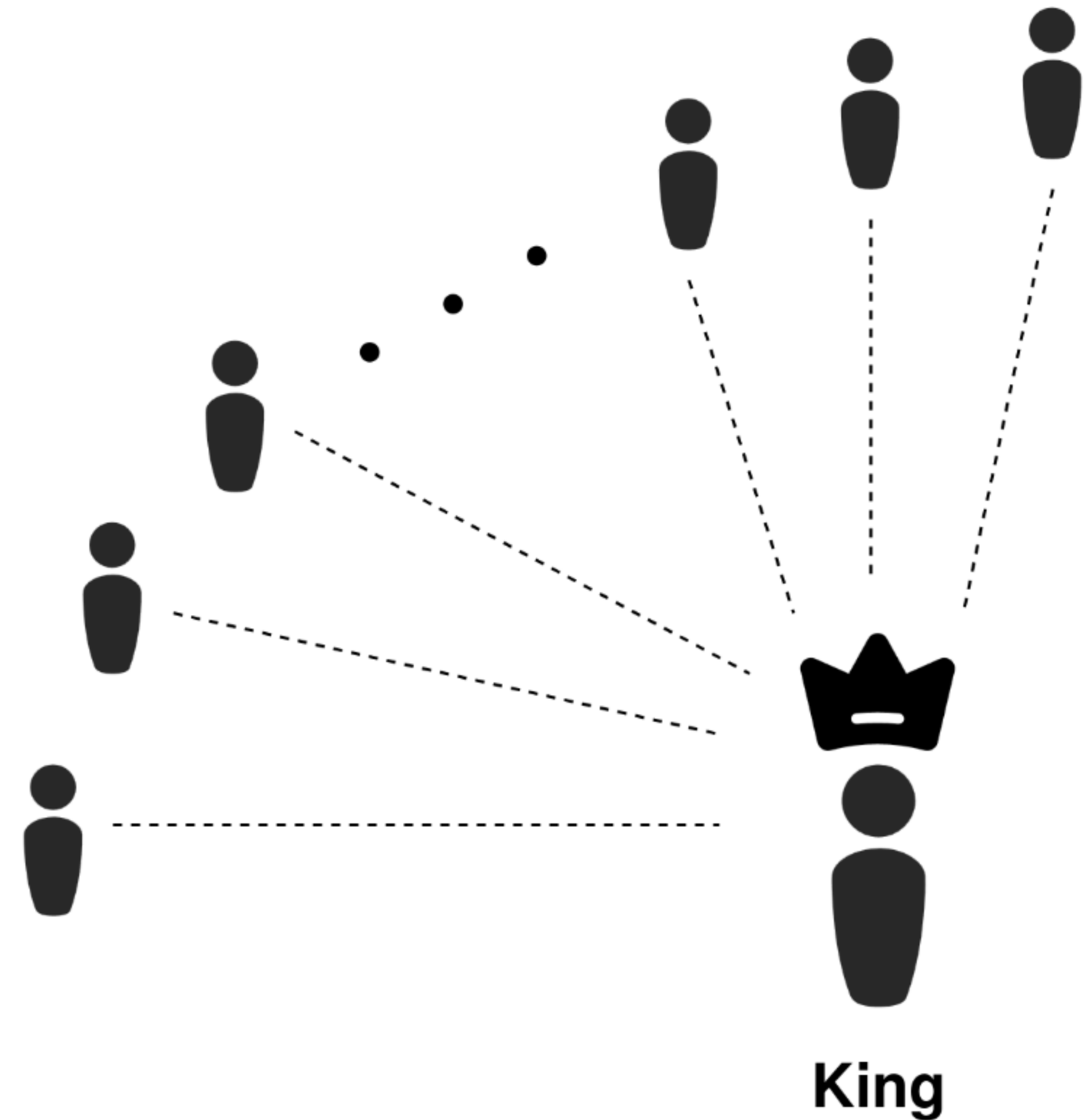
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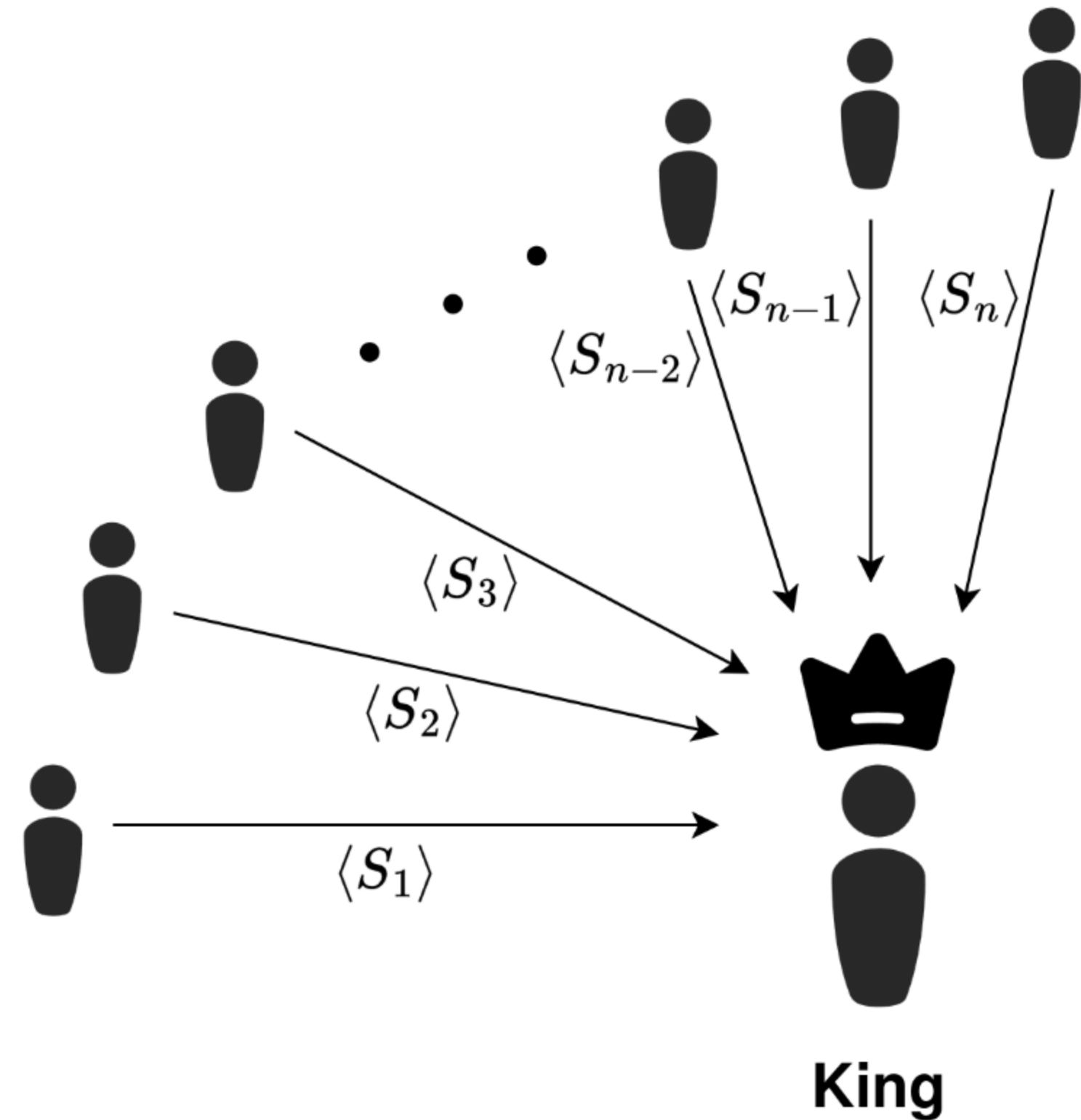
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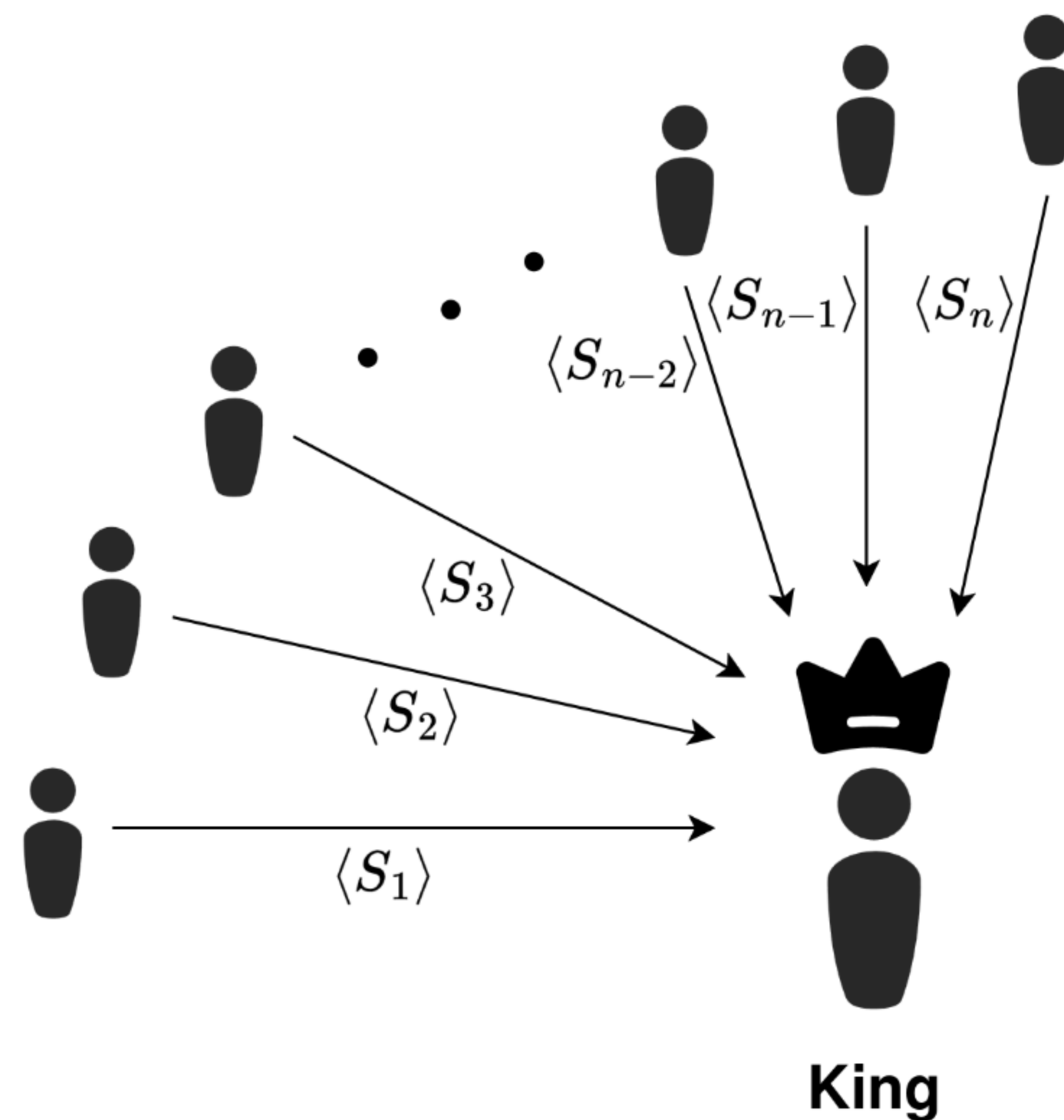
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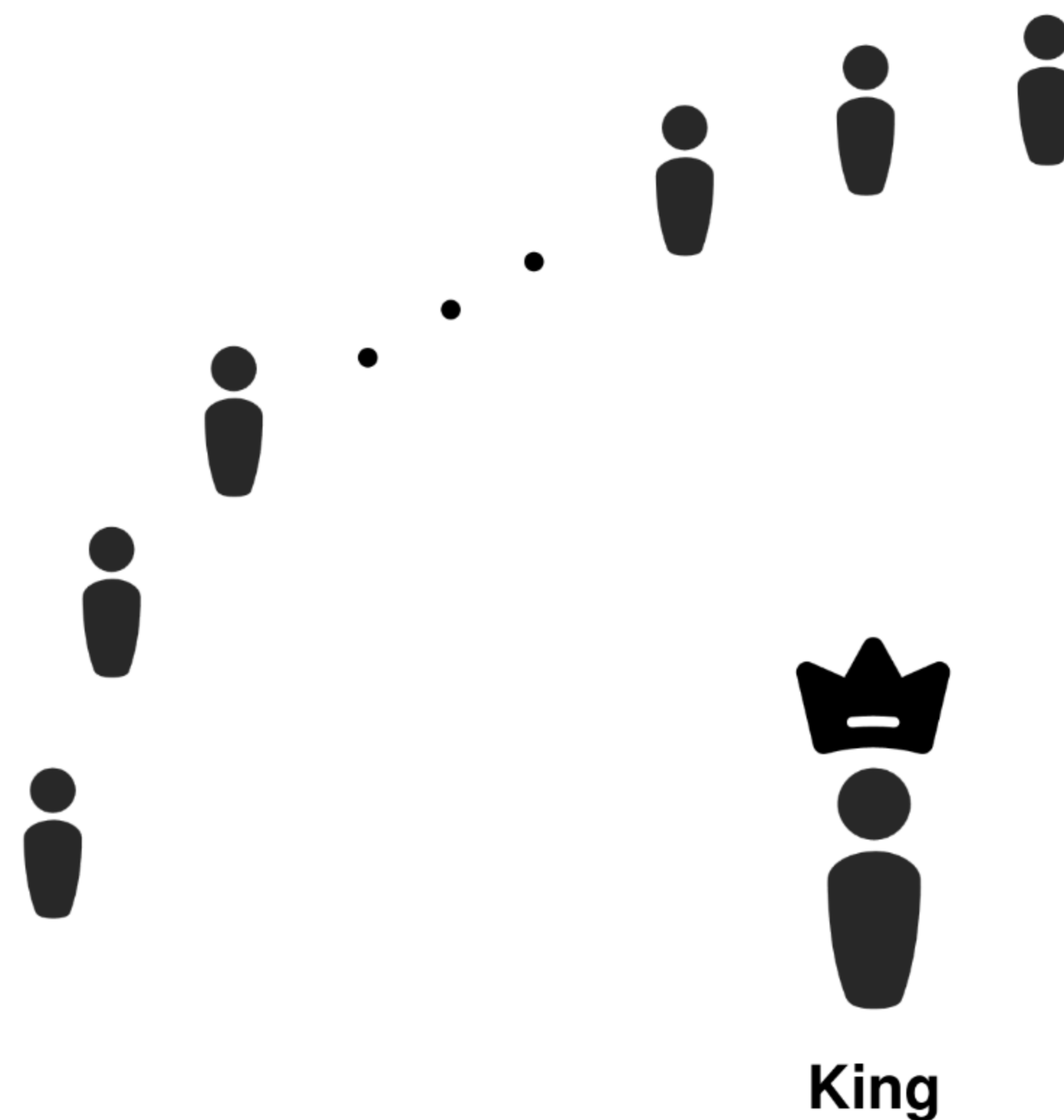
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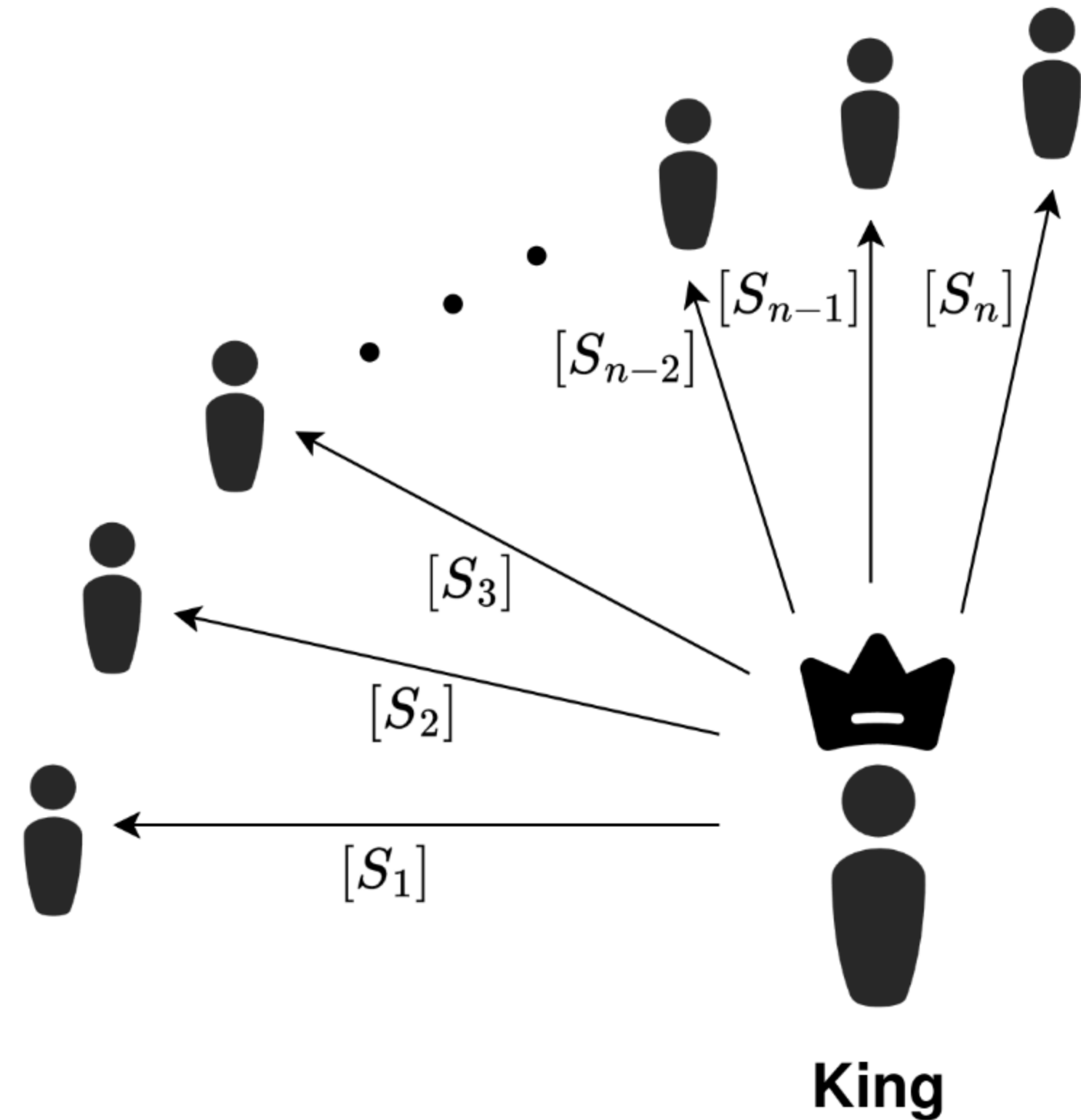
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Thank You!